J.P. Morgan digest on risk premia strategies

Summary of Q1’26 research reports on systematic investing

Cross Asset Systematic Strategy

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com
J.P. Morgan Securities plc

Thomas Salopek
(1-212) 834-5476
thomas.salopek@jpmorgan.com
J.P. Morgan Securities LLC

Yangyang Hou
(44-20) 3493-1012
yangyang.hou@jpmorgan.com
J.P. Morgan Securities plc

Mengdi Wang
mengdi.wang@jpmchase.com
J.P. Morgan Securities LLC

Meera Chandan
(44-20) 7134-2924
meera.chandan@jpmorgan.com
J.P. Morgan Securities plc

Antonin T Delair
(44-20) 7134-4643
antonin.t.delair@jpmorgan.com
J.P. Morgan Securities plc

Ioannis Pantelidakis, DPhil
(44-20) 3493 3625
ioannis.pantelidakis@jpmorgan.com
J.P. Morgan Securities plc

Mengjiao Wang
(44-78) 9704-9696
mengjiao.wang@jpmorgan.com
J.P. Morgan Securities plc

Sandra Djambazovska
sandra.djambazovska@jpmorgan.com
J.P. Morgan Securities plc

Robert Smith, PhD
(61-2) 9003-8808
robert.z.smith@jpmorgan.com
J.P. Morgan Securities Australia Limited

Chris Chi
chris.chi@jpmorgan.com
J.P. Morgan Securities Australia Limited

See page 74 for analyst certification and important disclosures.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Table Of Contents

J.P. Morgan Systematic Strategies Research Contacts

Quantitative Research

Dubravko Lakos-Bujas
dubravko.lakos-bujas@jpmorgan.com
J.P. Morgan Securities LLC

Khuram Chaudhry
khuram.chaudhry@jpmorgan.com
J.P. Morgan Securities plc

Thomas Salopek
thomas.salopek@jpmorgan.com
J.P. Morgan Securities LLC

Dobromir Tzotchev, PhD
dobromir.tzotchev@jpmorgan.com
J.P. Morgan Securities plc

Robert Smith, PhD
robert.z.smith@jpmorgan.com
J.P. Morgan Securities Australia Limited

Ayub Hanif, PhD
ayub.hanif@jpmorgan.com
J.P. Morgan Securities plc

William Summer
william.summer@jpmorgan.com
J.P. Morgan Securities plc

Arun Jain
arun.p.jain@jpmorgan.com
J.P. Morgan Securities LLC

Ioannis Pantelidakis, DPhil
ioannis.pantelidakis@jpmorgan.com
J.P. Morgan Securities plc

Sandra Djambazovska
sandra.djambazovska@jpmorgan.com
J.P. Morgan Securities plc

Mengjiao Wang
mengjiao.wang@jpmorgan.com
J.P. Morgan Securities plc

Chris Chi
chris.chi@jpmorgan.com
J.P. Morgan Securities Australia Limited

Equity Derivatives Strategy

Tony SK Lee
tony.sk.lee@jpmorgan.com
J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong)

Haoshun Liu
haoshun.liu@jpmorgan.com
J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong)

Davide Silvestrini
davide.silvestrini@jpmorgan.comJ.P. Morgan Securities plc

Yangyang Hou
yangyang.hou@jpmorgan.com
J.P. Morgan Securities plc

Esmail Afsah
esmail.afsah@jpmorgan.com
J.P. Morgan Securities plc

Mengdi Wang
mengdi.wang@jpmorgan.com
J.P. Morgan Securities LLC

Credit Derivative and Index Research

Saul Doctor
saul.doctor@jpmorgan.com
J.P. Morgan Securities plc

Global FX Strategy

Meera Chandan
meera.chandan@jpmorgan.com
J.P. Morgan Securities plc

Ladislav Jankovic
ladislav.jankovic@jpmorgan.com
J.P. Morgan Securities LLC

Arindam Sandilya
arindam.x.sandilya@jpmorgan.com
JPMorgan Chase Bank, N.A., Singapore Branch

Antonin T Delair
antonin.t.delair@jpmorgan.com
J.P. Morgan Securities plc

Global Macro Strategy

Nikolaos Panigirtzoglou
nikolaos.panigirtzoglou@jpmorgan.com
J.P. Morgan Securities plc

Mika Inkinen
Mika.j.inkinen@jpmorgan.com
J.P. Morgan Securities plc

Global Fixed Income and US Equity Index Technical Strategy

Jason Hunter
Jason.x.hunter@jpmorgan.com
J.P. Morgan Securities LLC

Emerging Markets Strategy

Jonny Goulden
jonathan.m.goulden@jpmorgan.com
J.P. Morgan Securities plc

Saad Siddiqui
saad.siddiqui@jpmorgan.com
J.P. Morgan Securities LLC

Rates Derivatives Strategy

Ipek Ozil
ipek.ozil@jpmorgan.com
J.P. Morgan Securities LLC

Philip Michaelides
philip.michaelides@jpmchase.com
J.P. Morgan Securities LLC

Arjun Parikh
arjun.parikh@jpmchase.com
J.P. Morgan Securities LLC

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Executive Summary

The current report is a part of periodic series that summarizes the publications of the J.P.Morgan research department that focus on risk premia investing and systematic strategies. Below we review the relevant research notes in Q1'26 organised in the following main topics:

1. Tactical risk premia research
    - Deep dive into the 2025 performance of the JPM Research risk premia strategies
    - QIS implementation of the 2026 cross-asset views
    - ARPs during geo-political stress
2. Delta-one systematic strategies
    - Deep Momentum
    - Optimized trend-following in factors
    - Rates Macro Quant: Forecast Revisions as a Measure of Macro Momentum
    - APAC Quant - A Deeper Look at Pair Trading
    - Enhanced trend-following overlay to equity quality
    - Profiting from the third moment in single stocks
    - Profiting from the Third Moment in Generic Equity Factors (GEFs)
3. Derivatives quant research and systematic strategies
    - What to Long, Short and When (Part II)
    - Systematic Options Strategies in VIX
    - Systematic Trades in Oil and Gold Options

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

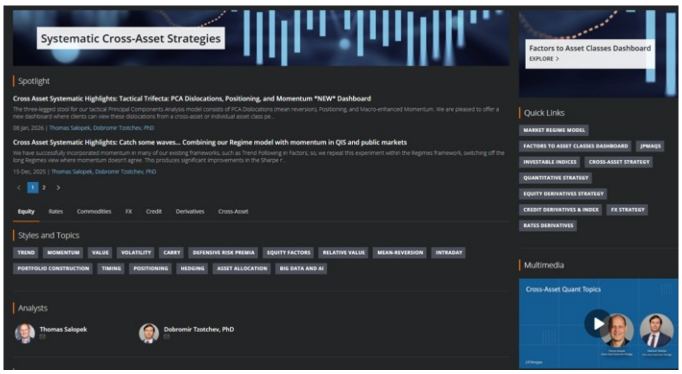
Asset Allocation and Tactical Risk Premia Research

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Deep dive into the 2025 performance of the JPM Research risk premia strategies

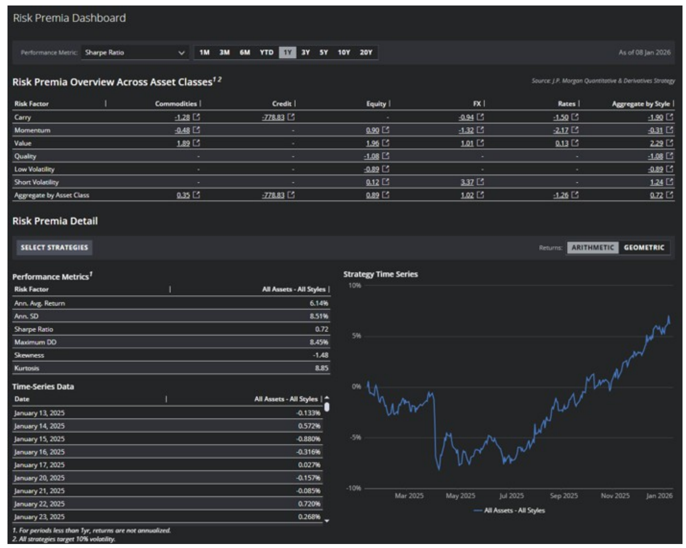
Our website 'Systematic Cross-Asset Strategies Research' on JPM Markets contains a wealth of functionality designed to allow users to quickly locate relevant publications on a topic and most importantly to track the performance of the latest generation of risk premia strategies (those implementations that we consider best in class) via interactive backtests and performance statistics.

Figure 1: Keyword publications locator on https://jpmm.com/#research.strategy.rbi



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 2: Performance dashboard on https://jpmm.com/#research.strategy.rbi



Source: J.P. Morgan Cross Asset Systematic Strategy

The table below summarizes strategies that we monitor once we group them via the familiar style/asset grid.

Table 1: Individual risk premia strategies components by style and asset class

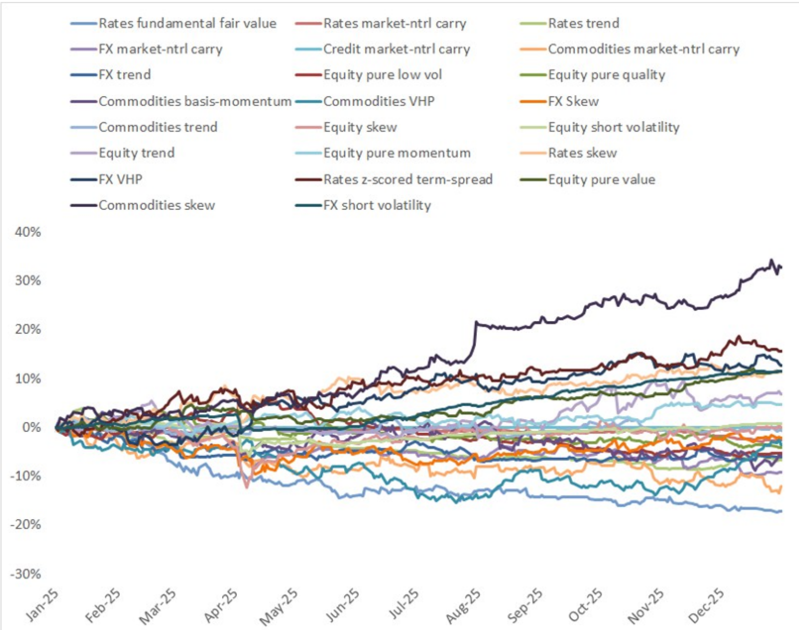
	Commodity	Credit	Currency	Equity	Covernment
Carry	Commodities market-ntrl carry	Credit market-ntrl carry	FX market-ntrl carry		Rates market-ntrl carry
Momentum	Commodities trend		FX trend	Equity pure momentum	Rates trend
	Commodities basis-momentum			Equity trend	
Value	Commodities VHP		FX Skew	Equity pure value	Rates z-scored term-spread
	Commodities skew		FX VHP	Equity skew	Rates fundamental fair value
					Rates skew
Low volatility				Equity pure low vol	
Short vol				Equity short volatility	
Quality				Equity pure quality	

Source: J.P. Morgan Cross Asset Systematic Strategy

In Deep dive into the 2025 performance of the JPM Research risk premia strategies we focus on the performance of the individual components in 2025.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 3: Performance by individual components



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 2: Performance statistics by individual components

strat	ann ret	ann std	sharpe	maxdd
Commodities skew	28.9%	10.1%	2.86	2.7%
Equity pure value	11.0%	4.8%	2.28	3.7%
Rates z-scored term-spread	15.0%	9.3%	1.60	3.9%
FX VHP	12.3%	9.6%	1.28	5.4%
Rates skew	11.3%	9.6%	1.17	7.3%
all assets - all styles	**6.5%**	**8.4%**	**0.78**	**8.5%**
Equity pure momentum	4.8%	6.8%	0.71	4.1%
Equity trend	7.1%	10.5%	0.68	7.1%
Equity short volatility	1.0%	5.7%	0.17	6.7%
Equity skew	0.5%	9.9%	0.05	14.4%
Commodities trend	-0.4%	5.8%	-0.07	4.0%
FX Skew	-1.7%	10.2%	-0.16	10.9%
Commodities VHP	-3.1%	10.6%	-0.29	15.3%
Commodities basis-momentum	-5.1%	13.3%	-0.38	11.6%
Equity pure quality	-4.1%	5.4%	-0.77	7.4%
Equity pure low vol	-5.3%	6.1%	-0.86	10.8%
FX trend	-6.1%	6.7%	-0.90	8.8%
Commodities market-ntrl carry	-12.4%	11.3%	-1.09	14.3%
Credit market-ntrl carry	0.0%	0.0%	-1.13	0.0%
FX market-ntrl carry	-9.4%	7.9%	-1.19	11.8%
Rates trend	-6.8%	5.3%	-1.27	12.1%
Rates market-ntrl carry	-2.6%	1.7%	-1.49	3.0%
Rates fundamental fair value	-18.5%	8.9%	-2.08	17.1%

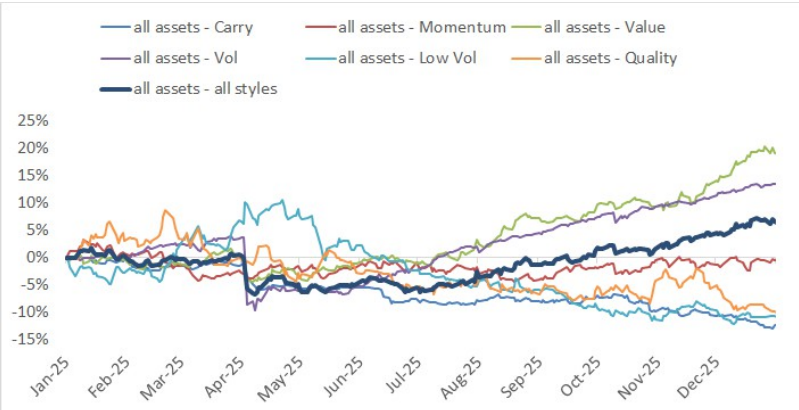
Source: J.P. Morgan Cross Asset Systematic Strategy

Skew strategies were winners in all asset classes, except for FX Skew which was marginally negative. Good performance from Skew offered support to the Value style, which was the strongest performer among the high-level style baskets in 2025 and also very strong in absolute performance terms, as we will see in Table 3. The notable outlier within the Value strats cohort was the Rates DFFV (Deviation From Fair Value) strategy which was the biggest laggard, wiping out last year's good performance. **Rates** strategies are clustered at the weakest end of the performance table, with the exception of Rates Skew, which bucked the general strength seen in the Skew signal this year; still, within our suite of strategies, Rates as an asset class were the worst performing. It was a challenging year for **Trend**, with Equities Trend the sole outperformer (and Equities Momentum along with it). Commodities Trend was only slightly negative, while FX and Rates Trend both lagged, in large part due to large and rapid shifts in interest rate expectations in the US.

The individual components are aggregated via a risk budgeting algorithm to arrive at the combined performance. A more detailed explanation of the methodology can be found here.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 4: Performance by style



Source: J.P. Morgan Cross Asset Systematic Strategy

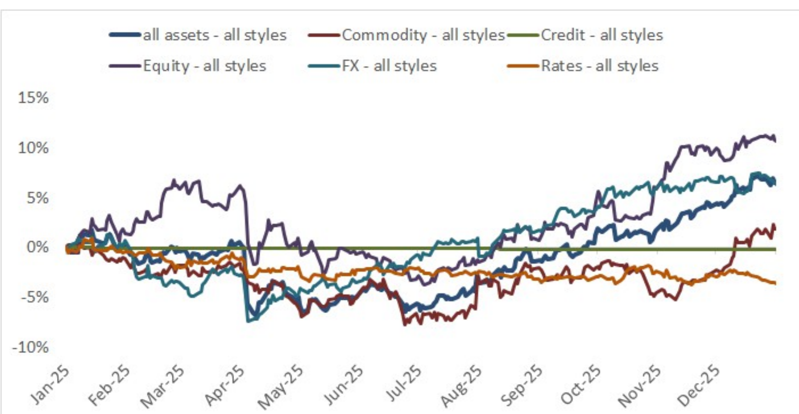
Table 3: Performance statistics by style

strat	ann ret	ann std	sharpe	maxdd
all assets - Value	17.8%	8.5%	2.09	6.0%
all assets - Vol	13.4%	11.6%	1.16	13.1%
all assets - all styles	**6.5%**	**8.4%**	**0.78**	**8.5%**
all assets - Momentum	-0.2%	8.4%	-0.02	7.7%
all assets - Quality	-9.7%	12.8%	-0.76	17.2%
all assets - Low Vol	-10.7%	12.4%	-0.87	20.5%
all assets - Carry	-13.1%	5.8%	-2.24	13.3%

Source: J.P. Morgan Cross Asset Systematic Strategy

In terms of performance of the main styles, this was another good year for **Short vol** and especially **Value**, both of which performed strongly and consistently since their short-lived Liberation Day selloff, which was especially sharp for Short Vol. On the flipside, the equity-specific **Quality** and **Low Vol** strategies were negative. **Carry** was the worst performer.

Figure 5: Performance by asset class



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 4: Performance statistics by asset class

strat	ann ret	ann std	sharpe	maxdd
Equity - all styles	10.7%	10.4%	1.02	9.8%
FX - all styles	6.5%	7.8%	0.84	8.5%
all assets - all styles	**6.5%**	**8.4%**	**0.78**	**8.5%**
Commodity - all styles	2.3%	9.1%	0.25	8.6%
Rates - all styles	-3.5%	4.2%	-0.84	4.7%
Credit - all styles	-0.2%	0.0%	-	0.2%

Source: J.P. Morgan Cross Asset Systematic Strategy

Turning our attention to performance by asset class, this was a year of broad-based, if choppy, strong performance, driven by the **Equities** and **FX** strategies which delivered another solid year. The aggregate all-strategy portfolio spent much of the year in the red, before staging a steady recovery in H2 and making it well into into positive territory in Q4. Indeed, notwithstanding the gyrations of 2025, the aggregated portfolio generated a **Sharpe ratio of over 0.7** through year end.

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

QIS implementation of the 2026 cross-asset views

Dobromir Tzotchev, PhD
dobromir.tzotchev@jpmorgan.com
J.P. Morgan Securities plc

Sandra Djambazovska
sandra.djambazovska@jpmorgan.com
J.P. Morgan Securities plc

In Implications of the 2026 outlooks for QIS investors we analyzed the potential impact of the J.P. Morgan 2026 outlook for risk premia strategies. Building on the market similarity framework as described in Risk premia sensitivity to rates/equities and Implications of JPM 2019 FICC outlooks and our latest timing forecasts, in QIS implementation of the 2026 cross-asset views we identify periods that are most similar to the conditions expected by the J.P. Morgan strategists. We use the Mahanalobis distance to determine the similarity between the forecasts and the realized values of these asset classes over 260-day rolling periods since December 2002. Given the higher importance of some asset classes/regions, we use a weighting scheme [1]. We then select the 5 non-overlapping 1-year periods that have conditions that are most like the J.P. Morgan Strategy Outlook (a conceptually similar approach is JP Morgan's Time Traveller).

In descending order of similarity, the 1-year periods ending September 2006, November 2004, May 2025, April 2011 and November 2023 are most alike the J.P. Morgan 2026 Outlook projections.

We expand our analysis of the 2026 outlook by focusing on practical QIS implementations. First, we present the results of the optimized delta-one portfolio. Here we take the expected returns from the J.P. Morgan 2026 outlook, construct a long-short basket of the delta-one instruments and evaluate the performance over the 5 most similar 1-year periods mentioned above. Next, we design an optimized portfolio of XRP strategies that maximizes the return in the most similar periods.

Optimized long-short basket of delta-one instruments

We investigate the construction of a long-short basket of delta-one instruments, leveraging end-of-year forecast levels for major markets across a broad range of asset classes. In addition to the level changes, we take into account dividends, carry, and the risk-free rate to arrive at the total expected return in 2026. We use those expected returns in combination with the covariance matrix to create an optimized long-short basket of those delta-one instruments.

The basket targets 10% ex-ante volatility and the weights are subject to maximum position sizes, as are the net exposures to asset classes and regions. Table 5 shows instruments under consideration, together with the total expected returns and the position limits. Positions and group limits have been chosen to represent the importance and/or underlying liquidity of the assets/groups and have been sized to ensure that the target volatility is achievable with reasonable leverage.

1. Effectively this means that we give more weight to Equities, Rates and Credit and less to Currency and Commodity and within asset classes more to the US.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Table 5: Investable universe and limits

ticker	name	asset class	region	expct ret	lower limit	upper limit
ES1 Index	S&P 500	Equity	US	6.75%	-80.4%	80.4%
VG1 Index	Eurostoxx 50	Equity	EU	7.43%	-15.4%	15.4%
Z 1 Index	FTSE 100	Equity	UK	5.97%	-3.7%	3.7%
TP1 Index	Topix	Equity	JP	11.77%	-5.9%	5.9%
MES1 Index	MSCI EM	Equity	EM	14.57%	-12.2%	12.2%
HI1 Index	Hang Seng	Equity	CN	18.84%	-2.5%	2.5%
JFBE3RXU Index	Germany 10yr	Rates	EU	1.54%	-29.9%	29.9%
JFBG3GUS Index	UK 10yr	Rates	UK	-1.53%	-7.0%	7.0%
JFBJ3JBU Index	Japan 10yr	Rates	JP	1.74%	-25.1%	25.1%
JFBU3TYU Index	US 10yr	Rates	US	-1.54%	-58.0%	58.0%
ERINCLIG Index	US IG	Credit	US	0.24%	-19.0%	19.0%
ERINCLHY Index	US HY	Credit	US	2.16%	-3.0%	3.0%
ERIXILIG Index	Europe IG	Credit	EU	1.19%	-7.1%	7.1%
ERIXILXO Index	Europe HY	Credit	EU	4.78%	-0.9%	0.9%
EURUSDCR Curncy	Euro	FX	EU	1.19%	-15.0%	15.0%
JPYUSDCR Curncy	Yen	FX	JP	-3.87%	-6.0%	6.0%
GBPUSDCR Curncy	Pound	FX	UK	3.34%	-3.0%	3.0%
CNYUSDCR Curncy	Yuan	FX	CN	-3.11%	-6.0%	6.0%
CL1 Comdty	WTI	Comdty	GL	-8.50%	-7.5%	7.5%
CO1 Comdty	Brent	Comdty	GL	-6.50%	-7.5%	7.5%
GC1 Comdty	Gold	Comdty	GL	12.70%	-15.0%	15.0%
Group Limits						
asset class	Comdty				-30.0%	30.0%
	Credit				-30.0%	30.0%
	Equity				-120.0%	120.0%
	FX				0.0%	0.0%
	Rates				-120.0%	120.0%
region	CN				-2.5%	2.5%
	EM				-12.2%	12.2%
	EU				-53.3%	53.3%
	GL				-30.0%	30.0%
	JP				-31.0%	31.0%
	UK				-10.7%	10.7%
	US				-160.4%	160.4%

Source: J.P. Morgan Cross Asset Systematic Strategy

At the beginning of each of the five most similar periods, we construct a portfolio by maximizing expected return while targeting 10% volatility. The expected returns used in each optimization are identical, based on the current forecasts. However, the covariance matrix is specific to each period, calculated from the 52 weeks of returns leading up to the start date of that period. Given that we aim to benefit from the forecasted asset returns, we don't impose delta neutrality – the baskets are allowed to take directional bets. The baskets are buy-and-hold, i.e. after the initial optimization at the beginning of the period, no further rebalancing takes place. All results are presented net of transaction and running costs and are in excess of the risk-free rate.

In Table 6 we examine the composition across the different periods. Although the expected returns remain constant, the covariance matrix changes, leading to variations in portfolio weights. As expected, the outlook pushes the portfolio towards riskier assets, with relatively large net positions in equities, while rates and credit assets are on average marginally underweighted. Within asset classes, higher expected returns translate to larger positions, subject to position and volatility constraints.

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Table 6: Long-short baskets of delta-one instruments

					start date					
ticker	**name**	**asset class**	**region**	**expected return**	**Nov-03**	**Sep-05**	**Apr-10**	**Nov-22**	**May-24**	***current***
ES1 Index	S&P 500	Equity	US	6.75%	68.5%	77.8%	40.6%	20.0%	60.2%	***46.0%***
VG1 Index	Eurostoxx 50	Equity	EU	7.43%	-15.4%	7.5%	1.0%	7.5%	15.4%	***7.4%***
Z 1 Index	FTSE 100	Equity	UK	5.97%	3.7%	3.7%	3.7%	3.7%	3.7%	***3.7%***
TP1 Index	Topix	Equity	JP	11.77%	5.9%	5.9%	5.9%	5.9%	5.9%	***5.9%***
MES1 Index	MSCI EM	Equity	EM	14.57%	0.0%	0.0%	12.2%	12.2%	12.2%	***12.2%***
HI1 Index	Hang Seng	Equity	CN	18.84%	2.5%	2.5%	2.5%	2.5%	2.5%	***2.5%***
net Equity exposure					*65.1%*	*97.3%*	*65.8%*	*51.7%*	*99.8%*	***77.7%***
JFBE3RXU Index	Germany 10yr	Rates	EU	1.54%	29.9%	29.9%	29.9%	29.9%	22.0%	***29.9%***
JFBG3GUS Index	UK 10yr	Rates	UK	-1.53%	-7.0%	-7.0%	-7.0%	-7.0%	-7.0%	***-7.0%***
JFBJ3JBU Index	Japan 10yr	Rates	JP	1.74%	25.1%	25.1%	25.1%	25.1%	25.1%	***25.1%***
JFBU3TYU Index	US 10yr	Rates	US	-1.54%	-42.6%	-57.0%	-58.0%	-58.0%	-58.0%	***-58.0%***
net Rates exposure					*5.4%*	*-9.0%*	*-10.0%*	*-10.0%*	*-17.9%*	***-10.0%***
ERINCLIG Index	US IG	Credit	US	0.24%			-19.0%	-19.0%	-19.0%	***-19.0%***
ERINCLHY Index	US HY	Credit	US	2.16%			-3.0%	-3.0%	-3.0%	***3.0%***
ERIXILIG Index	Europe IG	Credit	EU	1.19%			7.1%	7.1%	7.1%	***7.1%***
ERIXILXO Index	Europe HY	Credit	EU	4.78%			0.9%	0.9%	0.9%	***0.9%***
net Credit Exposure							*-14.0%*	*-14.0%*	*-14.0%*	***-8.0%***
EURUSDCR Curncy	Euro	FX	EU	1.19%	7.9%	7.9%	7.9%	7.9%	7.9%	***7.9%***
JPYUSDCR Curncy	Yen	FX	JP	-3.87%	-6.0%	-6.0%	-6.0%	-6.0%	-6.0%	***-6.0%***
GBPUSDCR Curncy	Pound	FX	UK	3.34%	3.0%	3.0%	3.0%	3.0%	3.0%	***3.0%***
CNYUSDCR Curncy	Yuan	FX	CN	-3.11%	-4.9%	-4.9%	-4.9%	-4.9%	-4.9%	***-4.9%***
net FX exposure					*0.0%*	*0.0%*	*0.0%*	*0.0%*	*0.0%*	***0.0%***
CL1 Comdty	WTI	Comdty	GL	-8.50%	-7.5%	-7.5%	-7.5%	-7.5%	-7.5%	***-7.5%***
CO1 Comdty	Brent	Comdty	GL	-6.50%	5.4%	4.0%	-7.5%	-4.6%	-7.5%	***-7.5%***
GC1 Comdty	Gold	Comdty	GL	12.70%	15.0%	15.0%	15.0%	15.0%	15.0%	***15.0%***
net Commodity exposure					*12.9%*	*11.5%*	*0.0%*	*2.9%*	*0.0%*	***0.0%***
net exposure					**83.4%**	**99.8%**	**41.8%**	**30.6%**	**67.9%**	***59.6%***
gross exposure					**263.0%**	**276.1%**	**267.6%**	**253.5%**	**293.7%**	***279.5%***

Source: J.P. Morgan Cross Asset Systematic Strategy

The performance of the optimized long-short delta-one baskets is evaluated across the five historical periods identified as most similar to the current JPM Research Outlook. Figure 6 and Table 7 show the results of the baskets under the five regimes:

Figure 6: Cumulative net performance in the 5 regimes

Source: J.P. Morgan Cross Asset Systematic Strategy

Table 7: Performance characteristics

	29/11/2004	28/09/2006	12/04/2011	08/11/2023	08/05/2025	all
sdate	28/11/2003	28/09/2005	12/04/2010	08/11/2022	08/05/2024	**28/11/2003**
edate	30/11/2004	29/09/2006	13/04/2011	09/11/2023	09/05/2025	**09/05/2025**
ann avg	9.4%	12.6%	3.3%	9.7%	11.5%	**9.3%**
ann std	8.1%	10.1%	11.5%	6.8%	15.0%	**10.7%**
sharpe	1.16	1.25	0.29	1.42	0.77	**0.87**
maxdd	5.3%	11.6%	8.8%	5.4%	15.0%	**15.0%**
skew	-0.26	-0.24	-0.23	-0.17	0.18	**-0.03**
kurt	0.27	1.12	1.17	0.92	9.38	**8.56**

Source: J.P. Morgan Cross Asset Systematic Strategy

The delta-one baskets perform well under the regimes we've identified. The Sharpe ratio over all the five 1-year periods is 0.87 and maximum drawdowns are contained given the basket's 10% target volatility. The basket can be used as an overlay on a traditional multi-asset portfolio, thereby boosting returns at desired risk levels. This can be done without a large capital allocation requirement as all the instruments used are forward/futures based.

Optimized XRP portfolio

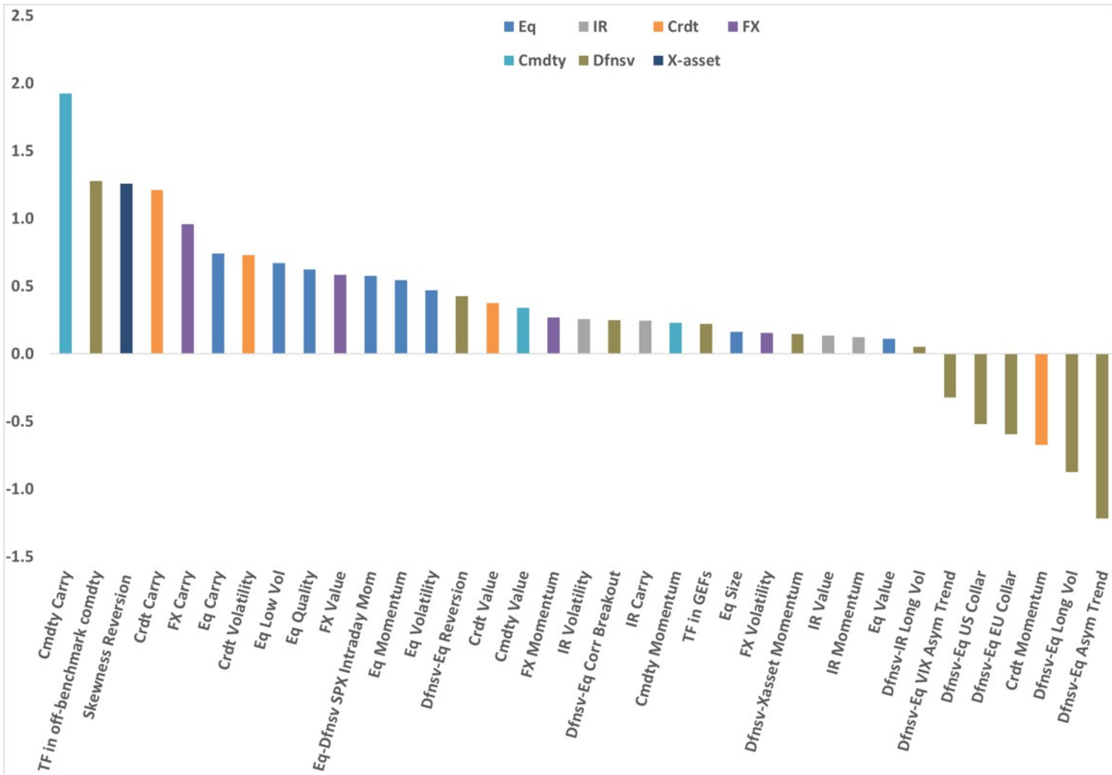
In Implications of the 2026 outlooks for QIS investors, we already identified which cross-asset risk premia strategies performed well in the regimes most similar to the one from the 2026 outlook (Figure 7 below).

Of course, individual strategies don't make a portfolio, so this month we turn our attention to that aspect. The portfolio weights are determined using an optimization :

- We maximize expected return, subject to 10% volatility. The expected return is the strategy average return across the five most similar regimes.
- The covariance matrix is estimated using the last three years of daily data.
- Each strategy's weight is bounded between zero and three times its risk-parity (ERC) weight.
- Additionally, we impose the constrain that the expected return in each of the five most similar periods should be positive.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

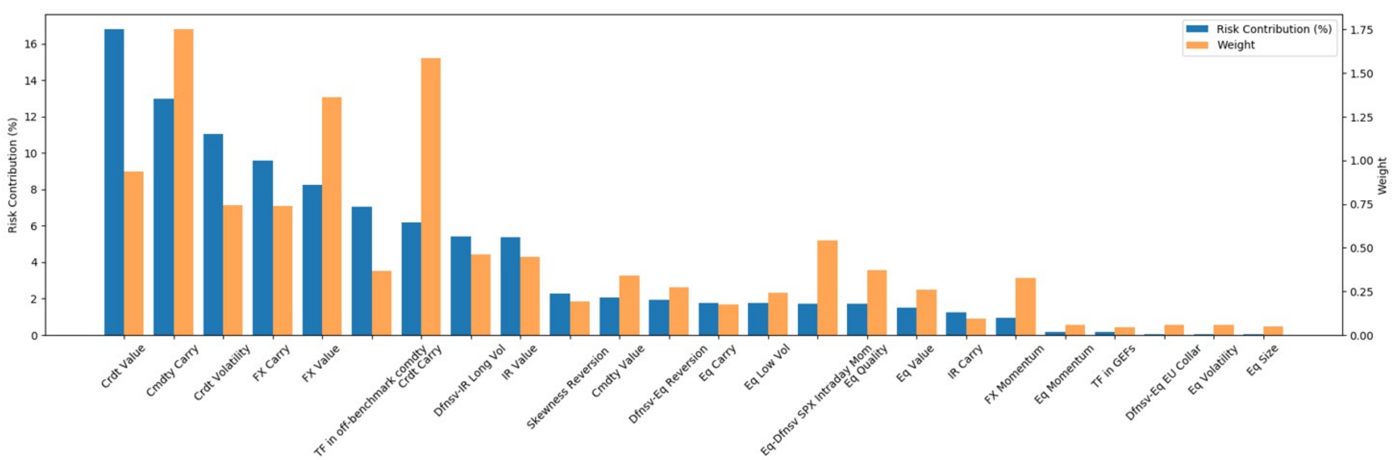
Figure 7: Sharpe ratios of the cross-asset risk premia strategies under the forecasted regime



Source: J.P. Morgan Cross Asset Systematic Strategy

Below we also show the actual optimized weights as well the risk contribution of the invidual strategies. Given that we require the portfolio return to be positive in any of the most similar periods, important is not only the Sharpe ratio of the strategies in the five regimes, but also the consistency of the performance across the regimes and the diversification benefits.

Figure 8: Allocated weights and risk contributions



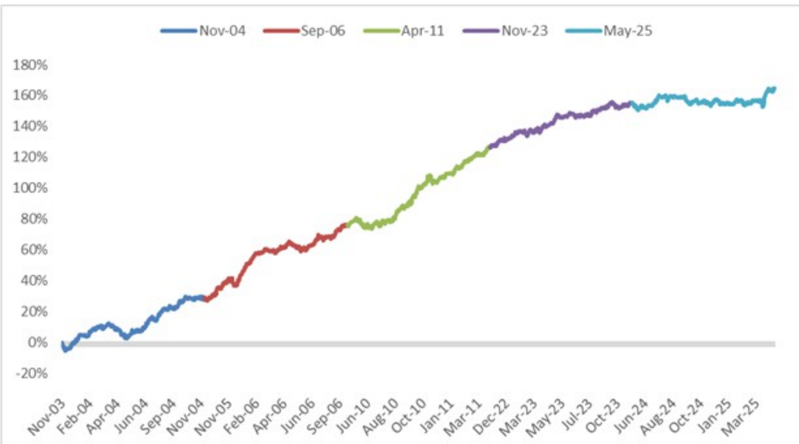
Source: J.P. Morgan Cross Asset Systematic Strategy

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Credit Value, Commodity Carry, Short Credit Vol, FX Carry and FX Value are the top 5 risk contributors to the optimized portfolio.

We first present the performance in the regime periods. The optimized portfolio achieves a Sharpe ratio of 2.87 across the selected regimes. The period ending May'25 (this inlcudes the market volatility around Liberation Day) is found to be the most challenging - nevertheless, the Sharpe ratio in that period is 0.85. The maximum drawdowns are also modest for the 10% target volatility.

Figure 9: Cumulative net performance XRP portfolio under regimes



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 10: Performance characteristics

	Sep-06	Nov-04	May-25	Apr-11	Nov-23	all
sdate	2005-09-28	2003-11-28	2024-05-08	2010-04-12	2022-11-08	**2003-11-28**
edate	2006-09-28	2004-11-29	2025-05-08	2011-04-12	2023-11-08	**2025-05-08**
ann avg	47.0%	28.8%	9.2%	50.0%	28.2%	**32.6%**
ann std	11.2%	11.0%	10.9%	13.0%	10.5%	**11.4%**
sharpe	4.20	2.63	0.85	3.86	2.67	**2.87**
maxdd	6.3%	9.4%	7.4%	7.0%	4.3%	**9.4%**
skew	-0.50	-0.22	-0.10	-0.17	-0.22	**-0.21**
kurt	1.60	0.93	3.55	0.86	0.69	**1.45**

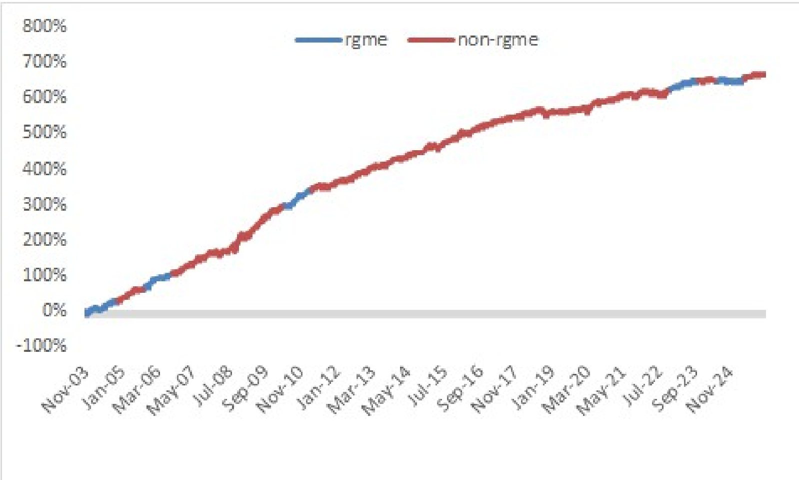
Source: J.P. Morgan Cross Asset Systematic Strategy

Also, given that these cross-asset risk premia strategies shouldn't be overly dependent on main asset class returns, we also present the results of the portfolio in the periods outside the five identified regimes.

The ERC combination of cross-asset risk premia provides excellent returns not only in the 5 regime years, but actually does well across the board. In the full sample though, the maximum drawdown is relatively large for the 10% target volatility.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 11: Cumulative net performance XRP portfolio full sample



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 8: Performance characteristics

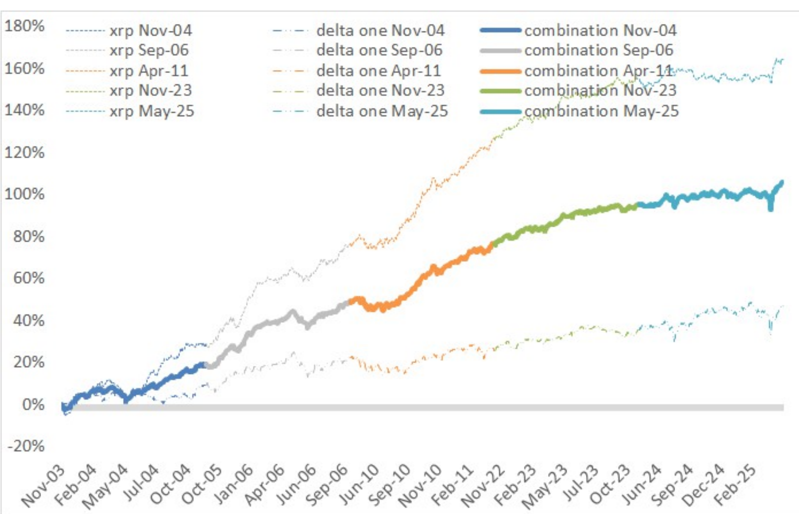
	rgme	non-rgme	all
sdate	2003-11-28	2004-11-29	**1997-01-01**
edate	2025-05-08	2026-01-06	**2026-01-06**
ann avg	32.6%	29.4%	**23.0%**
ann std	11.4%	15.2%	**12.6%**
sharpe	2.87	1.93	**1.82**
maxdd	9.4%	20.6%	**20.6%**
skew	-0.21	0.40	**0.49**
kurt	1.45	10.22	**13.98**

Source: J.P. Morgan Cross Asset Systematic Strategy

Combining the delta-one basket and the cross-asset risk premia portfolio

In this last part of the section, we evaluate the performance of the combination of the delta-one baskets and the cross-asset risk premia portfolio. Since both target 10% volatility, we simply allocate one half of the portfolio to the delta-one basket and the other half to the cross-asset risk premia portfolio.

Figure 12: Cumulative net performance combined portfolio in the regimes periods



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 13: Performance characteristics

	xrp	delta one	comb
ann avg ret	32.6%	9.3%	21.0%
ann std	11.4%	10.7%	8.5%
sharpe	2.87	0.87	2.48
maxdd	9.4%	15.0%	9.2%
skew	-0.21	-0.03	-0.24
kurt	1.45	8.52	2.99

Source: J.P. Morgan Cross Asset Systematic Strategy

The combination achieves a respectable Sharpe ratio of 2.48 over the selected regimes periods. This combination of the delta-one basket and xrp portfolio can be easily implemented as an overlay on a traditional multi-asset portfolio.

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

ARPs during geo-political stress

Dobromir Tzotchev, PhD
dobromir.tzotchev@jpmorgan.com
J.P. Morgan Securities plc

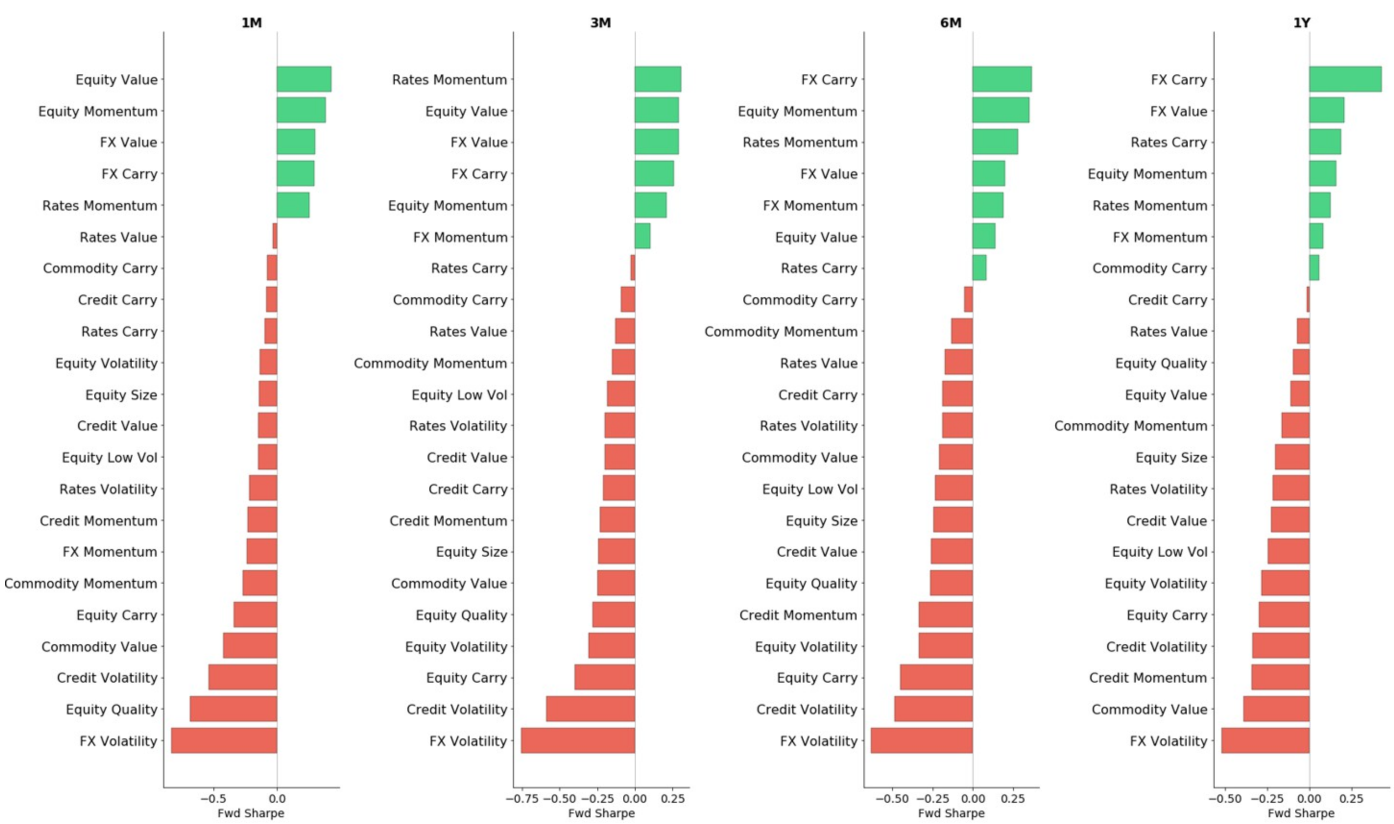
Sandra Djambazovska
sandra.djambazovska@jpmorgan.com
J.P. Morgan Securities plc

In ARPs during geo-political stress we examine how risk premia strategies perform in the periods following extreme geopolitical environments. We rely on the Geopolitical Risk Index (GPR) developed by Dario Caldara and Matteo Iacoviello in "Measuring Geopolitical Risk" (American Economic Review, April, 112(4), pp.1194-1225).

Using the Geopolitical Risk Index, we identify periods falling in the top quintile (Geopolitical Stress regime) or the bottom quintile (Geopolitical Calm regime). For each strategy, we compute the forward 1-month, 3-months, 6-months and 1-year as the ratio of the average annualized excess return above the long-term mean and the annualized historical strategy volatility. Then for each interval we compute the average of those forward scaled excess performance ratios.

In Figure 14,we rank all risk premia strategies by their average forward excess performance ratios, when the geopolitical risk index is in the top 20th percentile, with each panel corresponding to a different forward horizon.

Figure 14: Forward Excess Performance Ratios in Stress Regimes (Top 20% GRP)



Source: J.P. Morgan Cross Asset Systematic Strategy

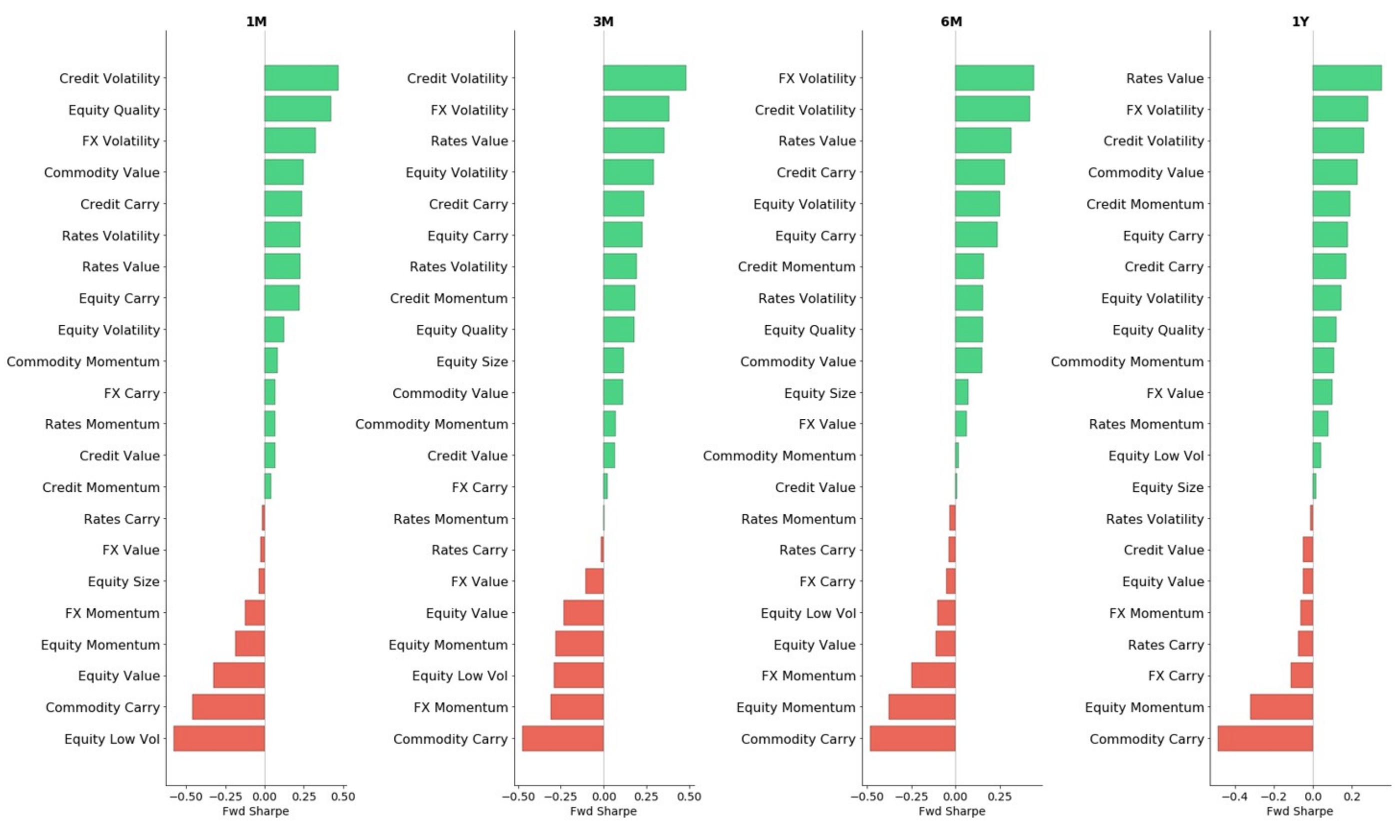
Several patterns emerge consistently across horizons. Short volatility strategies (in particular FX and Credit Volatility) are among the worst performers across all forward horizons. In contrast, momentum strategies and FX Value tend to do well. Somewhat counterintuitively, FX Carry is a strong performer as well which can potentially be

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

attributed to the strengthening of the commodity currencies. The results are relatively consistent among different forward horizons, which suggests that geo-political risks tend to cluster.

Figure 15, shows the corresponding analysis for calm periods, when the geopolitical risk index is in the bottom 20^{th} percentile. The results show a reversal in the pattern observed in stress periods. Short volatility strategies are now among the best performing strategies, while the momentum performance is more muted. Equity Momentum underperforms in calm periods. Equity Low Vol is the worst performer at the 1-month horizon, but the effect fades at longer horizons. Note that Equity Low Vol underperforms in periods of stress as well, which is consistent with our analysis that Equity Low Vol while a defensive style is short convexity.

Figure 15: Forward Excess Performance Ratios in Calm Regimes (Bottom 20% GPR)



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 16, provides a side-by-side comparison of the forward excess performance ratios in calm vs. stress regimes for each strategy, organized by asset class and style. The most pronounced differentiation appears in Volatility strategies, which exhibit positive forward ratios following calm periods and negative following stress periods across all asset classes and time horizons. Momentum strategies display the opposite pattern, tending to be positive during stress times and negative during calm periods. Similarly to Equity Low Vol, Commodity Carry also exhibits negative convexity with more pronounced underperformance in the calm period.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 16: Calm vs. Stress Regimes by Asset Class and Style

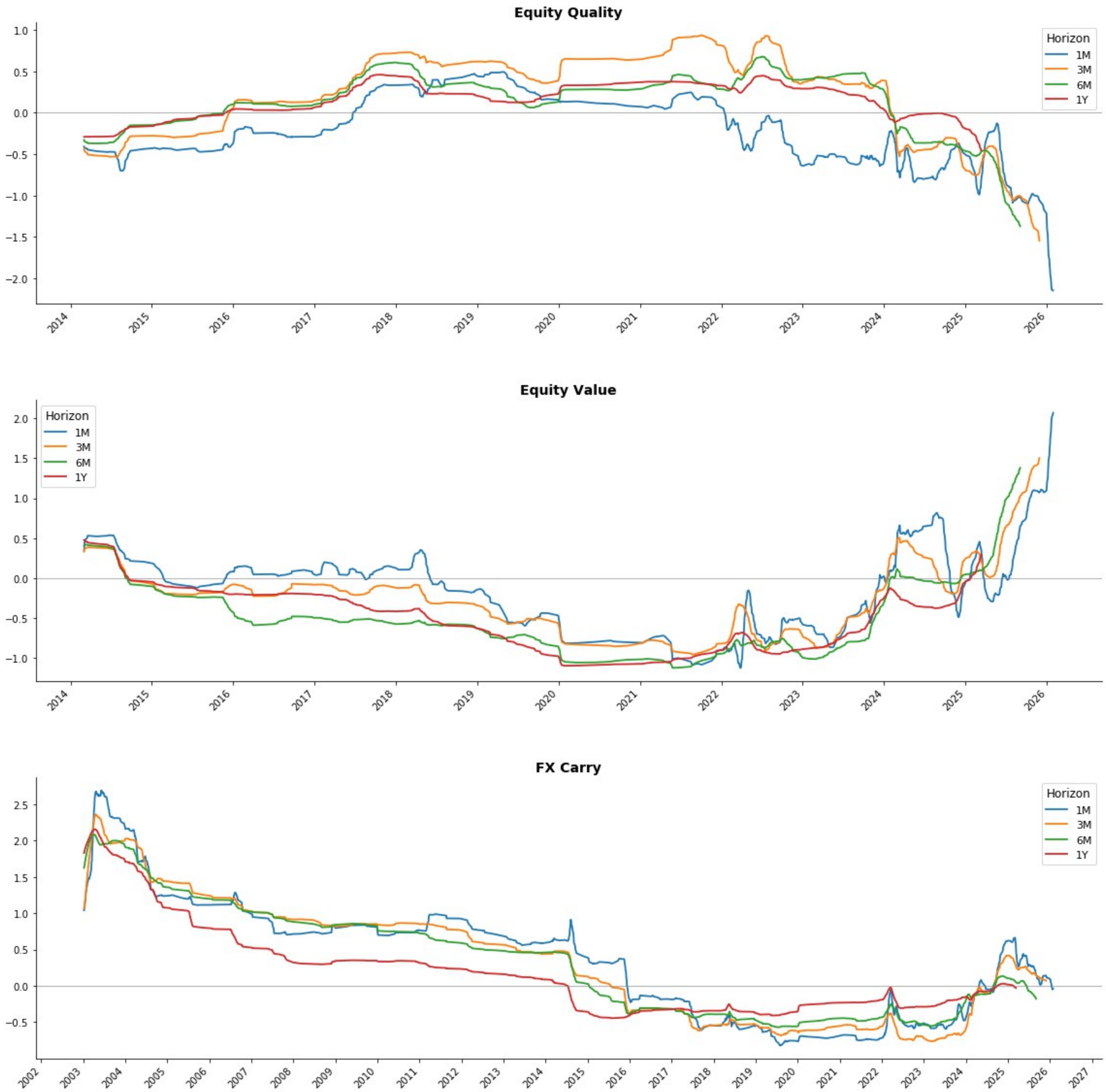
Average annualized forward Sharpe ratios

Source: J.P. Morgan Cross Asset Systematic Strategy

In Figure 17, we show the 252-observation rolling average of the forward excess return performance ratio across different horizons in the geopolitical stress regime, for Equity Quality, Equity Value and FX Carry. Equity Quality has had a mostly positive performance ratio across all horizons until 2024, but the average drops sharply after that. For Equity Value, we have the opposite picture: the ratio tends to be negative in geopolitical stress regimes until the last couple of years when it moves into positive territory. FX Carry had a positive ratio prior to 2015 and in recent years.

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 17: Rolling average of the forward excess return performance ratio



Source: J.P. Morgan Cross Asset Systematic Strategy

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Delta-1 Systematic Strategies

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Deep Momentum

Thomas Salopek
thomas.salopek@jpmorgan.com
J.P. Morgan Securities LLC

Mengjiao Wang
mengjiao.wang@jpmorgan.com
J.P. Morgan Securities plc

Dobromir Tzotchev, PhD
dobromir.tzotchev@jpmorgan.com
J.P. Morgan Securities plc

The delta-of-the-straddle momentum signal is designed to capture both the strength and the uncertainty of trend estimates. Statistically, it is equivalent to testing whether the mean return of an asset is significantly positive or negative over a given period. With appropriately chosen strike and maturity, the signal can be interpreted as the delta of a straddle option. As the signal value falls within the range of [-1, 1], it takes into account the empirical fact that outsized returns are less probable.

Our standard trend-following framework, introduced in Designing robust trend-following system, falls within the time series momentum category, focusing on each asset's own historical returns—buying assets with strong past performance and selling those with weaker trends. Specifically, we compute and average the delta-of-the-straddle signal across multiple lookback horizons (32 days, 3 months, 6 months, 1 year, and 2 years). The position in each asset is set proportional to its aggregated signal and inversely proportional to its volatility. Let $X^{(i)}_t$ denote the aggregated delta-of-the-straddle signal of asset i at time t. If all assets receive equal risk weight, we can model the strategy return as:

$$\frac{S}{N}\sum_{i=1}^{N}\frac{X_t^{(i)}}{\sigma_t^{(i)}}r_{t,t+1}^{(i)}$$

where S is a scaler to target the portfolio at 10% annualized vol on an expanding window.

Rationale of Our Study

While standard trend-following momentum strategies have proven effective across asset classes, they typically rely on signals derived solely from each asset's own historical performance, without accounting for the relative behavior or interactions among assets within the portfolio. Additionally, traditional momentum signals are generally agnostic to the optimal holding period—they indicate the presence of a trend but do not provide guidance on how long the signal remains predictive or actionable.

In Deep Momentum Models, we use deep neural networks to predict optimal portfolio allocations on the cross-asset universe, using delta-of-the-straddle momentum signals as inputs. By employing a custom loss function that directly maximizes the portfolio Sharpe ratio, our approach ensures that position sizing is explicitly optimized for risk-adjusted performance at the portfolio level. Moreover, since the Sharpe ratio is computed using future returns over the intended holding period, the resulting positions are inherently aligned with the timeframe during which the signal is expected to be effective. We implement both Pooled Asset-wise and Cross-Asset Deep Momentum Model architectures, enabling the framework to flexibly capture both individual asset trends and cross-sectional momentum dynamics.

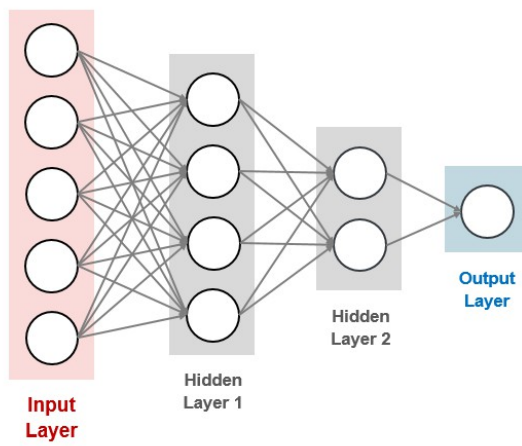
Model Foundations

Deep learning is a branch of machine learning that leverages multi-layered neural networks to automatically learn complex patterns and representations from large datasets. Among the most fundamental architectures in deep learning is the multi-layer perceptron (MLP), which consists of interconnected layers of artificial neurons that

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

transform input features through a series of weighted sums and nonlinear activation functions. In the context of finance, MLPs are particularly powerful for modeling intricate relationships among market variables, forecasting asset returns, and optimizing portfolio allocations. By capturing nonlinear dependencies and interactions that traditional linear models may overlook, deep learning models such as MLPs can enhance predictive accuracy and support more sophisticated decision-making in areas like credit scoring, fraud detection, and algorithm trading. An example of MLP is shown below.

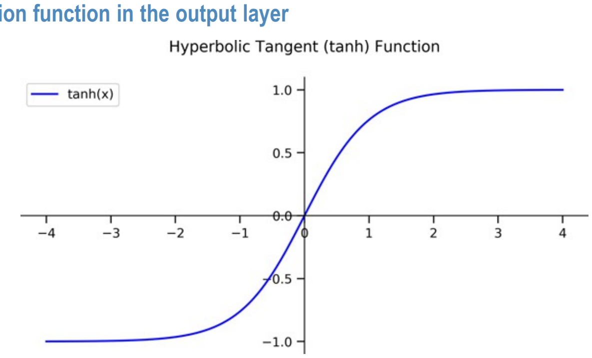
Figure 18: Example of a Multi-Layer Perceptron with 2 hidden layers, 5 input variables and 1 output variable



Source: J.P. Morgan Research

In this study, we fix the portfolio rebalancing frequency to once per month, with all models retrained at the end of each month using the most recent 10 years of daily samples. Prior to training, features with more than 30% missing values are removed, and any remaining missing values are imputed with zeros to maintain data integrity. For each hidden layer, batch normalization is applied prior to the ReLU activation function to stabilize and accelerate training. The output variable, in this case the optimal asset position is bounded within the range of [-1, 1] by a hyperbolic tangent activation function.

Figure 19: Activation function in the output layer



Source: J.P. Morgan Research

Let $X^{(i)}_t$ denote the estimated position for asset i at time t, and let c represent the transaction cost. The strategy's net return from opening and holding this position over

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

the following month is given by:

$$R_{t,t+1} = \frac{S}{N}\sum_{i=1}^{N}\left(\frac{X_t^{(i)}}{\sigma_t^{(i)}} r_{t,t+1}^{(i)} - c\left|\frac{X_t^{(i)}}{\sigma_t^{(i)}}\right|\right)$$

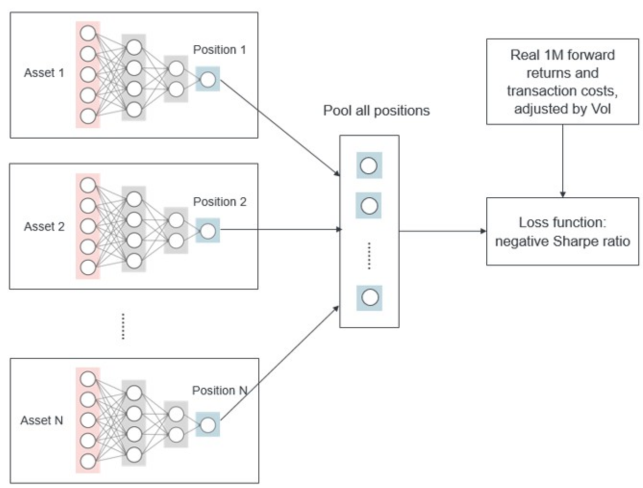
where $r^{(i)}_{t,\, t+1}$ is the one-month forward return for asset i. Our objective function for training the model is directly defined as the strategy Sharpe ratio, which is a function of the strategy net returns:

$$\text{Loss function} = -\frac{E(R_t)}{\sigma(R_t)}$$

Pooled Asset-wise Deep Momentum Model

In this framework, a separate multi-layer perceptron is trained for each asset using only that asset's own delta-of-the-straddle momentum features (32 days, 64 days, 126 days, 252 days, and 504 days) as input. Each model follows the same architecture as in Figure 20 and outputs the optimal position for its respective asset. These positions are then pooled to construct the portfolio. Models are trained to minimize the loss function computed on the pooled portfolio. Please find the visualisation below to better understand the process.

Figure 20: Overview of the pooled asset-wise deep momentum framework

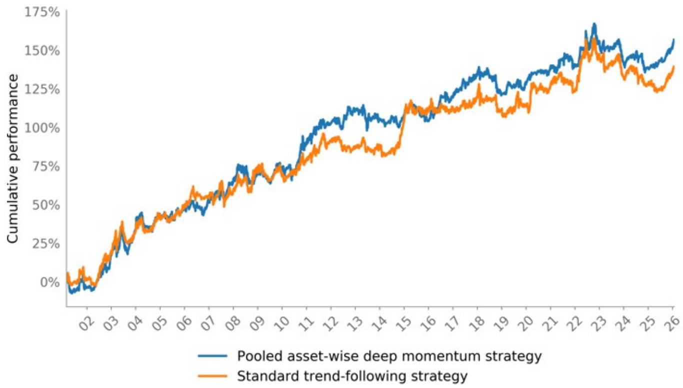


Source: J.P. Morgan Research

Model training is conducted using mini-batches of 64 samples, optimizing the network parameters via stochastic gradient descent with the Adam optimizer. The learning rate is set to 0.001, and L2 regularization (weight decay) with a coefficient of 0.001 is applied to all trainable parameters to mitigate overfitting. Early stopping is employed to terminate training if no improvement is observed for 10 consecutive epochs. To further enhance the stability and robustness of the model's predictions, an ensemble approach is adopted: the model is independently trained using 10 different random seeds, and the final prediction for each asset is obtained by averaging the outputs of these 10 outputs.

Below we compare the performance of the pooled asset-wise deep momentum and the standard trend-following, applying to the same investment universe. For both approach, we rebalance the portfolio once per month and rescale the portfolio to achieve 10% annualized volatility on expanding window. The backtest embeds realistic transaction costs and 1 day lag in trades execution has also been incorporated.

Figure 21: Cumulative performance of the pooled asset-wise deep momentum and the standard trend-following strategy



Source: J.P. Morgan Research

Table 9: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Pooled asset-wise deep momentum strategy	6.08%	9.58%	0.63	-27.88%
Standard trend-following strategy	5.40%	10.11%	0.53	-30.45%

Source: J.P. Morgan Research

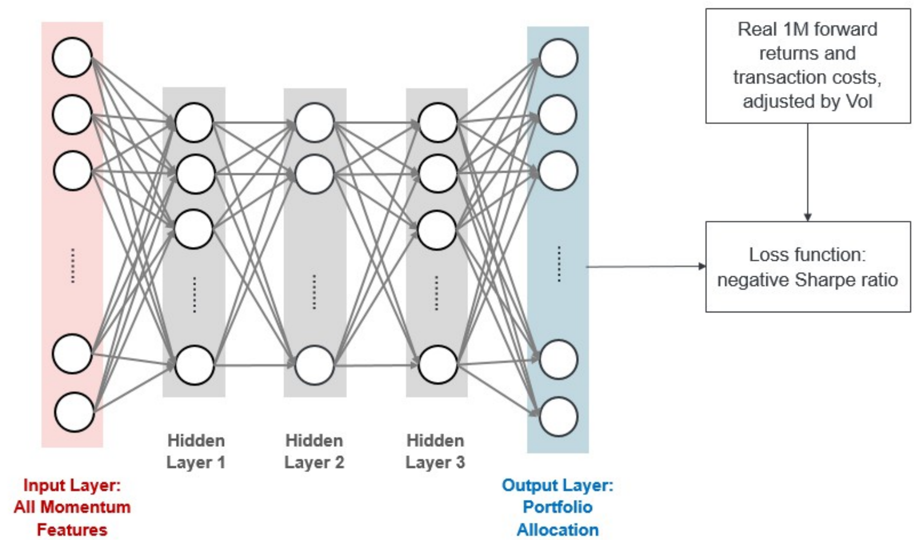
Cross-Asset Deep Momentum Model

In contrast to the pooled asset-wise framework, we also experiment with a single multi-layer perceptron that ingests the momentum features from all assets simultaneously and outputs a vector of positions for the entire asset universe. In this setup, each asset's position is determined not only by its own features but also by the features of all other assets, allowing the model to capture cross-sectional relationships and dependencies. This enables the cross-asset MLP to potentially exploit inter-asset information and learn more sophisticated allocation rules that may enhance portfolio-level risk-adjusted returns.

We start with a 3-layer MLP with hidden size [20, 10, 20]. The model is designed to first compress the cross-asset momentum features into a lower-dimensional latent space that captures the most informative aspects of the data, and then expand this representation to reconstruct the portfolio weights. A visualization of the process is displayed in Figure 22 .

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 22: Overview of the cross-asset deep momentum framework



Source: J.P. Morgan Research

Similarly, the model parameters are trained in mini-batches of 64 samples using the Adam optimizer. The learning rate is set to 0.01, and an L2 regularization weight of 0.001 is applied to all parameters to prevent overfitting. Early stopping is employed to terminate training if no improvement in the loss function is observed for 10 consecutive epochs. Again, the model is independently trained using 10 different random seeds, and the final prediction is the average of these 10 outputs.

Below we present the cumulative performance of the cross-asset deep momentum strategy. It exhibits strong long-term performance, with a Sharpe ratio of 0.76 over more than 20 years of backtest.

Figure 23: Cumulative performance of the cross-asset deep momentum and the standard trend-following strategy

Source: J.P. Morgan Research

Table 10: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Cross-asset deep momentum strategy	8.52%	11.16%	0.76	-25.20%
Standard trend-following strategy	5.40%	10.11%	0.53	-30.45%

Source: J.P. Morgan Research

Optimized trend-following in factors within and across asset classes

Dobromir Tzotchev, PhD
dobromir.tzotchev@jpmorgan.com
J.P. Morgan Securities plc

Mengjiao Wang
mengjiao.wang@jpmorgan.com
J.P. Morgan Securities plc

Rationale and methodology

In Risk premia performance review, trend-following in factors and latest model views, we presented and applied two optimized variants of the trend-following factors approach to commodities. The first implementation minimizes the cost (rebalancing and replication) of the strategy while maintaining the overall trend-following exposure. The second optimization maximizes the net expected return of the strategy (expected trend return minus the costs mentioned above). Both variants control deviations from the original trend-following in factors portfolio through tracking error constraints and limits on maximum deviations in individual positions. In Revisting Trend-Following in Factors Approach, we extended the application to all asset classes.

In Optimized trend-following in factors , we propose a new implementation that balances net trend return maximization with improved cost control. We then analyze the performance of this new strategy both within and across asset classes.

The portfolio is rebalanced weekly, every Wednesday, with positions held for the following week (5 business days). The expected weekly return for each asset is estimated as sqrt(5) * $S_{i,t}$* $Vol_{i,t}$, where $S_{i,t}$ is the delta-of-the-straddle trend-following signal introduced in Designing robust trend-following system and $Vol_{i,t}$ is the volatility of asset i at time t.

Recognizing that expected return estimates are inherently uncertain while costs are incurred with certainty, we quantify the estimation uncertainty of the expected returns and incorporate it as a penalty to the cost term. For each asset, we run an OLS regression of the one-week ahead realized return on our estimated expected return to obtain the estimated β. Empirically, β tends to be a small positive value less than one, indicating that our forecasts may overestimate expected returns by a factor of 1/β. On rare occasions the return forecasted direction might be incorrect and β might be negative. To address this, we cap 1/β within the range of 0 to 100 and use the average value as a penalty on the cost terms in the objective:

$$\text{Maximize } \sum_{i=1}^{N} w_{i,t} * \sqrt{5} * S_{i,t} * Vol_{i,t} - \overline{1/\beta_t} * \sum_{i=1}^{N} |w_{i,t}| * 5 * rc_i - \overline{1/\beta_t} * \sum_{i=1}^{N} |w_{i,t} - w_{i,t-1}| * tc_i$$

where rc_i is the daily running cost and tc_i is the transaction cost of asset i.

The additional constraints are linked to the standard trend-following in factors portfolio:

- The tracking error is limited to 25% of the portfolio volatility.
- The expected trend-return of the optimized portfolio shall be greater than the one of the trend-following in factors portfolio
- The portfolio volatility of the optimized portfolio shall be smaller or equal to the one of the standard trend-following in factors portfolio
- We impose additional position constraints as well to control the overall leverage of the portfolio and the deviation from the positions of the standard trend-following in factors portfolio:

$$0 \leq w_{opt} \leq 2 * w_{pca} \text{ if } w_{pca} > 0$$

$$2 * w_{pca} \leq w_{opt} \leq 0 \text{ if } w_{pca} < 0$$

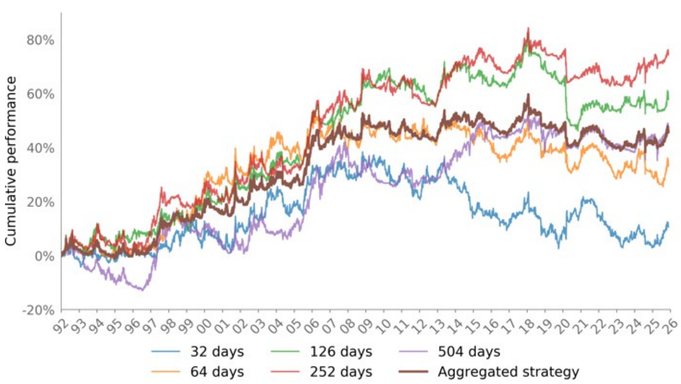
Empirical results

Below, we present the performance of the new optimized trend-following in statistical factors strategy by asset class across various lookback windows.

Overall, the optimized trend-following in factors strategy demonstrates positive performance within each asset class, achieving robust Sharpe ratios around 0.5 in rates and commodities, and approximately 0.3 in equities and currencies. As we have often observed, longer lookback periods (in particular one year) tend to deliver stronger results.

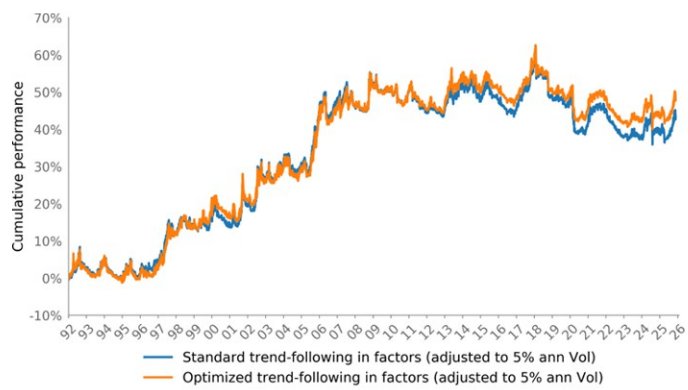
Furthermore, the optimized trend-following approach outperforms the standard version in general, with the most significant improvement observed in rates.

Figure 24: Cumulative performance of equities optimized trend-following in statistical factors across different lookback periods and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 25: Cumulative performance of equities standard and optimized trend-following in statistical factors



Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
32 days	0.30%	6.08%	0.05	-31.77%
64 days	0.94%	6.15%	0.15	-25.68%
126 days	1.64%	5.86%	0.28	-30.76%
252 days	2.10%	5.85%	0.36	-20.30%
504 days	1.32%	5.89%	0.22	-18.56%
Aggregated strategy	1.29%	4.77%	0.27	-19.87%

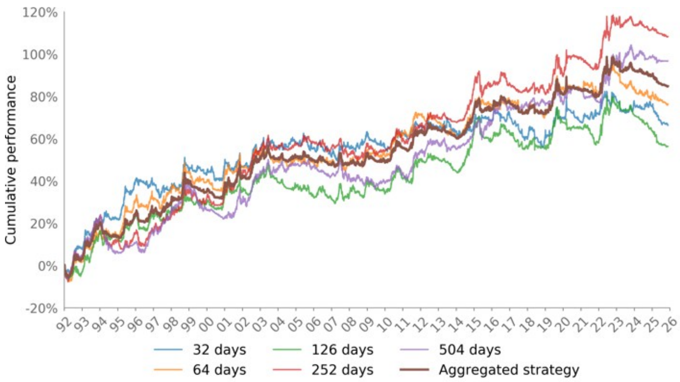
Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Standard trend-following in factors (adjusted to 5% ann Vol)	1.41%	5.82%	0.24	-26.67%
Optimized trend-following in factors (adjusted to 5% ann Vol)	1.29%	4.77%	0.27	-19.87%

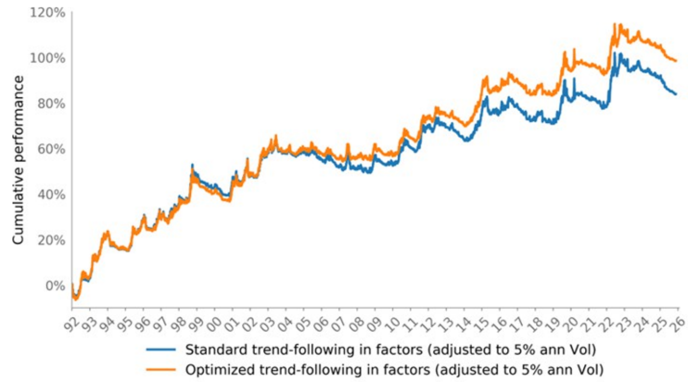
Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 26: Cumulative performance of rates optimized trend-following in statistical factors across different lookback periods and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 27: Cumulative performance of rates standard and optimized trend-following in statistical factors



Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
32 days	1.86%	5.24%	0.36	-17.65%
64 days	2.15%	5.18%	0.41	-20.02%
126 days	1.58%	5.11%	0.31	-22.04%
252 days	3.04%	5.22%	0.58	-15.02%
504 days	2.72%	5.39%	0.50	-18.30%
Aggregated strategy	2.38%	4.30%	0.55	-13.36%

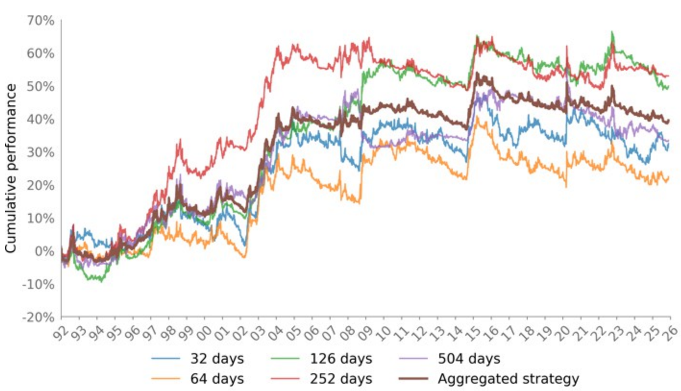
Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Standard trend-following in factors (adjusted to 5% ann Vol)	2.36%	5.00%	0.47	-17.02%
Optimized trend-following in factors (adjusted to 5% ann Vol)	2.77%	5.00%	0.55	-15.40%

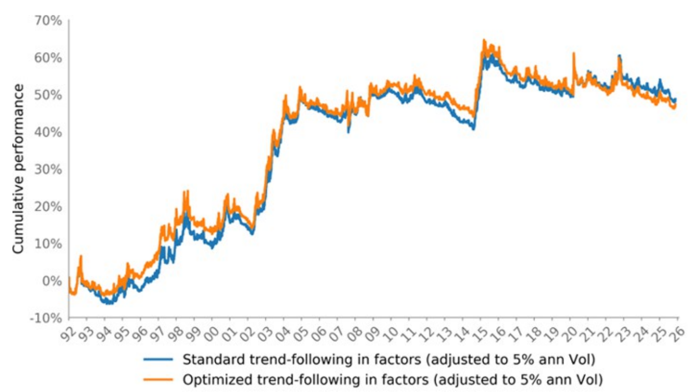
Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 28: Cumulative performance of FX optimized trend-following in statistical factors across different lookback periods and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 29: Cumulative performance of FX standard and optimized trend-following in statistical factors



Source: J.P. Morgan Cross Asset Systematic Strategy

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
32 days	0.91%	5.40%	0.17	-19.94%
64 days	0.63%	5.15%	0.12	-19.99%
126 days	1.40%	4.82%	0.29	-16.43%
252 days	1.49%	5.07%	0.29	-15.60%
504 days	0.94%	5.06%	0.19	-20.69%
Aggregated strategy	1.11%	4.18%	0.27	-14.97%

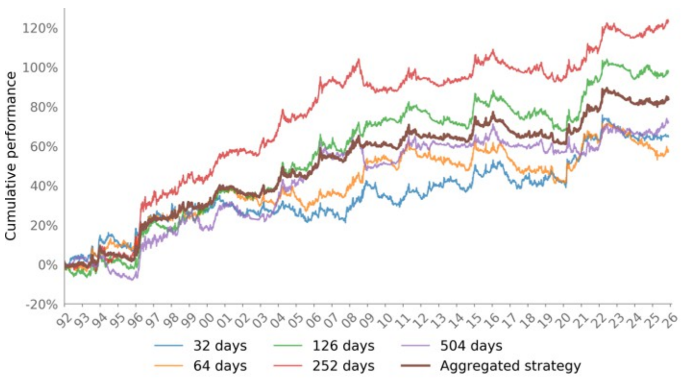
Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Standard trend-following in factors (adjusted to 5% ann Vol)	1.37%	5.00%	0.27	-13.72%
Optimized trend-following in factors (adjusted to 5% ann Vol)	1.33%	5.00%	0.27	-17.79%

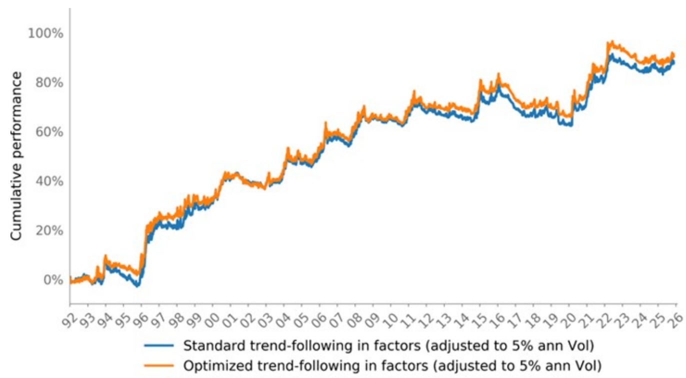
Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 30: Cumulative performance of commodities optimized trend-following in statistical factors across different lookback periods and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 31: Cumulative performance of commodities standard and optimized trend-following in statistical factors



Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
32 days	1.82%	5.72%	0.32	-13.87%
64 days	1.60%	5.60%	0.29	-20.83%
126 days	2.74%	5.55%	0.49	-19.04%
252 days	3.46%	5.67%	0.61	-16.18%
504 days	2.02%	5.53%	0.37	-16.56%
Aggregated strategy	2.35%	4.63%	0.51	-15.32%

Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Standard trend-following in factors (adjusted to 5% ann Vol)	2.46%	5.00%	0.49	-15.85%
Optimized trend-following in factors (adjusted to 5% ann Vol)	2.54%	5.00%	0.51	-16.44%

Source: J.P. Morgan Cross Asset Systematic Strategy

We also present results from applying these approaches to the entire universe of assets. In this methodology, PCA analysis is conducted on the full cross-asset universe. The aggregated strategy achieves a Sharpe ratio of approximately 0.7, with longer lookback periods consistently delivering superior performance.

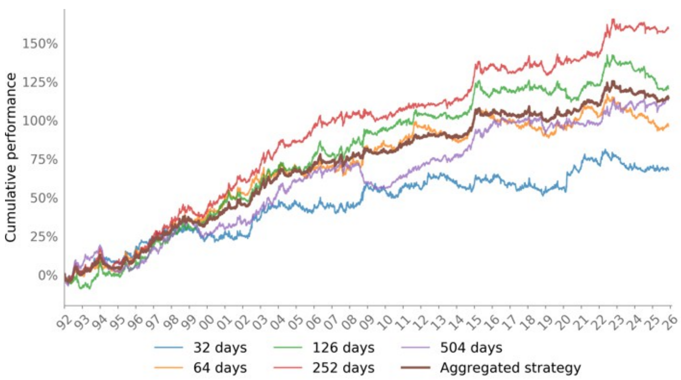
When comparing the trend-following in factors system to the conventional trend-following system as described in Designing robust trend-following system, the conventional approach has slightly outperformed over the backtest period. However, the merits of the trend-following in factors approach should not be evaluated solely on performance metrics, as discussed in Risk premia performance review, trend-following

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

in factors and latest model views. Notably, trend-following in factors offers the advantage of lower correlation to other investment styles. Also, the performance of trend-following in factors and the market-neutral cross-sectional momentum strategies shall be analyzed as a combination versus the conventional trend-following approach.

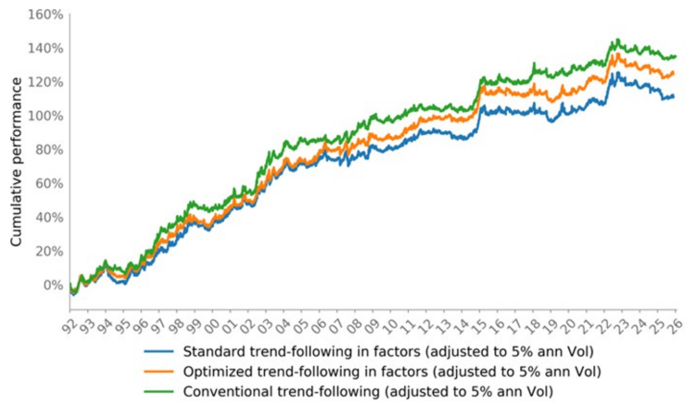
An additional benefit of the trend-following in factors methodology is that it eliminates the need for a single-asset risk weighting input, as this decision is internalized by the algorithm. Furthermore, the optimized version allows for the incorporation of additional constraints—such as maximum position size and maximum leverage—which could be important from a risk management perspective.

Figure 32: Cumulative performance of the optimized trend-following in cross-asset statistical factors across different lookback periods and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 33: Cumulative performance of standard and optimized trend-following in cross-asset statistical factors vs conventional trend-following



Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
32 days	1.91%	6.05%	0.32	-14.81%
64 days	2.71%	5.75%	0.47	-21.49%
126 days	3.38%	5.55%	0.61	-20.91%
252 days	4.48%	5.62%	0.80	-12.60%
504 days	3.18%	5.42%	0.59	-17.22%
Aggregated strategy	3.21%	4.59%	0.70	-12.72%

Source: J.P. Morgan Cross Asset Systematic Strategy

Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Standard trend-following in factors (adjusted to 5% ann Vol)	3.11%	5.00%	0.62	-15.08%
Optimized trend-following in factors ((adjusted to 5% ann Vol))	3.50%	5.00%	0.70	-13.79%
Conventional trend-following ((adjusted to 5% ann Vol))	3.77%	5.00%	0.75	-11.56%

Source: J.P. Morgan Cross Asset Systematic Strategy

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Rates Macro Quant: Forecast Revisions as a Measure of Macro Momentum

Antonin T Delair
antonin.t.delair@jpmorgan.com
J.P. Morgan Securities plc

Meera Chandan
meera.chandan@jpmorgan.com
J.P. Morgan Securities plc

In a similar vein to our previous work in FX, the note Rates Macro Quant: Forecast Revisions as a Measure of Macro Momentum explores in detail inflation and growth FRIs for systematic cross-market rates trading in DM.

In DM countries with a strong macro momentum, i.e. growth and inflation pushing higher together should see their rates market underperform compared to peers. In Rates Macro Quant: Forecast Revisions as a Measure of Macro Momentum we test whether economists' forecast revisions can be used to reflect this macro momentum and inform cross-market rates trading in DM.

Just as we have done for FX, we leverage our economists' Forecast Revision Indices for real GDP and headline CPI. Both are indices incremented by %-pts change in JPM growth and inflation forecasts (over a rolling 4 quarters period).

Our preferred signal is the average of growth and inflation 1y Z-Score of FRIs 3m change, which outperforms growth or inflation alone. Pooled regressions show a negative relationship between this composite and subsequent duration returns, though the relationship weakens after controlling for various effects.

Backtesting this signal on DM 10y bond futures and G10 5y swaps (top/bottom 2; monthly rebalanced) leads to a steady upward trajectory in the past 20+ years. Risk-adjusted returns are 0.38 for bond futures and 0.25 for swaps after trading costs (0.49 and 0.57, respectively, before costs). Trading costs have a significant impact on swaps (hence monthly vs. weekly rebalancing).

The FRI macro momentum strategy shows little to no consistent correlation with common FX and rates macro factors or key market variables, and has outperformed rates carry strategies, particularly post-Covid. The only significant correlation is with a similar macro momentum strategy applied to G10 FX. The FX implementation generally outperforms swaps post trading costs, but the upward trajectory is less consistent across time.

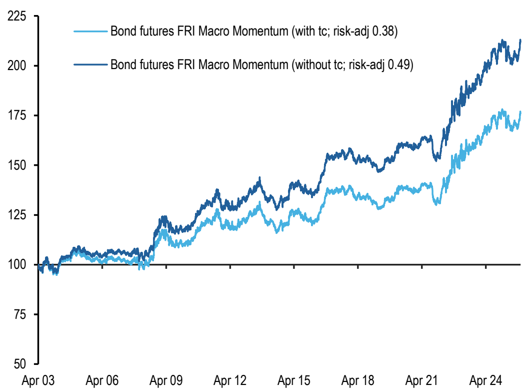
Overall, the FRI composite macro momentum is a robust feature for cross-market rates trading, with diversification benefits.

The following charts and tables summarize the empirical results.

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 34: A composite growth + inflation FRI momentum signal leads to a steady upward trajectory on 10y bond futures...

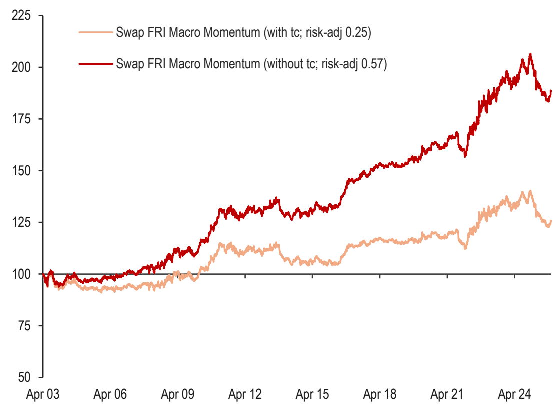
Total return index of L/S top 2/bot 2 DV01 neutral macro momentum strategy on 10Y bond futures (monthly rebalanced). Macro momentum is computed using an average of GDP and CPI FRIs momentum.



6 10Y Bond Futures: AU, CN, JB, UK, US, DE
Source: J.P. Morgan

Figure 35: ... and on G10 swaps as well; albeit more impacted by trading costs

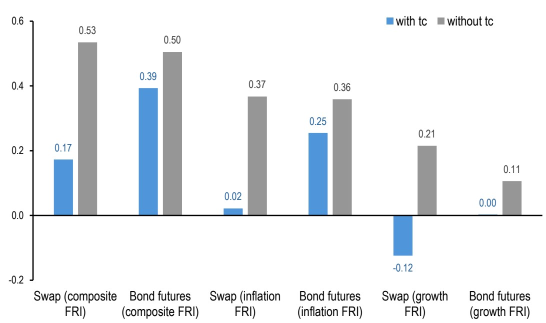
Total return index of L/S top 2/bot 2 DV01 neutral macro momentum strategy on 5Y swaps (monthly rebalanced). Macro momentum is computed using an average of GDP and CPI FRIs momentum.



G9 swaps: AUD, EUR, GBP, NZD, JPY, USD, GBP, NOK, SEK
Source: J.P. Morgan

Figure 36: Since 2013, the composite macro momentum has generally outperformed growth and inflation alone for bond futures and swaps

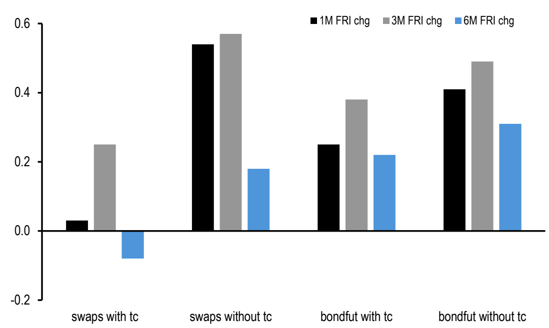
Risk-adjusted returns since 2013 of growth, inflation and composite FRI momentum strategies applied to bond futures and 5Y swaps baskets (top/bottom 2 monthly rebalanced).



6 10Y Bond Futures: AU, CN, JB, UK, US, DE
G9 5Y swaps: AUD, EUR, GBP, NZD, JPY, USD, GBP, NOK, SEK
Source: J.P. Morgan

Figure 37: 1M and 3M changes in FRI are more relevant for macro momentum features in rates

Risk-adjusted returns since 2003 of composite macro momentum strategy using inflation and growth FRIs (L/S top/bottom 2). Variation on FRI change window.



6 10Y Bond Futures: AU, CN, JB, UK, US, DE
G9 5Y swaps: AUD, EUR, GBP, NZD, JPY, USD, GBP, NOK, SEK
Source: J.P. Morgan

Figure 38: The FRI macro momentum strategy does not show any significant correlation with other factors or key market variables, except for a similar macro momentum built on G10 currencies

Correlation matrix of swap and bond futures FRI macro momentum strategies with other macro factors and key market variables. Period is 2003-2025 and the sampling is biweekly returns.

	Swap Macro Momentum	Bondfut Macro Momentum	Swap Carry	Bondfut Carry	Swap Value	Bondfut Value	G10 FX Carry	G10 FX Value	G10 FX Growth Momentum	G10 FX Macro Momentum	S&P 500	USD TWI
Bondfut Macro Momentum	0.51											
Swap Carry	-0.05	-0.09										
Bondfut Carry	-0.01	-0.11	0.66									
Swap Value	-0.08	0.07	-0.21	-0.17								
Bondfut Value	0.04	0.09	-0.14	-0.17	0.60							
G10 FX Carry	0.02	-0.06	0.09	0.12	-0.34	-0.22						
G10 FX Value	-0.06	0.03	-0.12	-0.12	0.20	0.17	-0.68					
G10 FX Growth Momentum	0.19	0.12	0.03	0.04	-0.05	-0.05	0.04	-0.11				
G10 FX Macro Momentum	0.25	0.15	0.04	0.04	-0.06	-0.03	0.06	-0.06	0.81			
S&P 500	-0.06	-0.03	-0.01	0.00	-0.10	-0.12	0.48	-0.22	0.02	0.08		
USD TWI	0.09	0.10	-0.10	-0.13	0.03	-0.03	-0.29	0.11	0.02	-0.03	-0.49	
10Y US yield	0.09	0.06	-0.22	-0.24	-0.23	-0.40	0.23	-0.12	-0.06	-0.06	0.17	0.11

For value:
In FX, we consider REER PPI deviation cross-sectional portfolios with high/low rich/cheap
In rates, we consider real yields cross-sectional portfolios with high/low real yields cheap/rich (defined as rate level - 5Y ahead inflation expectations).
Source: J.P. Morgan

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Implementing Pairs in Asia (ex Japan) with Market Constraints

Robert Smith
robert.z.smith@jpmorgan.com
J.P. Morgan Securities Australia Limited

Chris Chi
chris.chi@jpmorgan.com
J.P. Morgan Securities Australia Limited

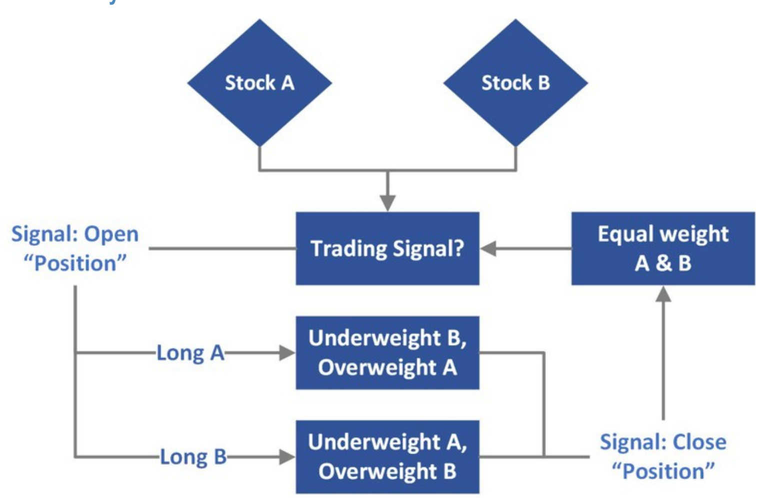
Pair trading, one of the oldest and most classic statistical arbitrage strategies, is predicated on the mean reversion of price spreads between two correlated assets.

We recently published a global note on a modernized pair trading model that fuses AI-driven insights to enhance traditional strategies. This strong base-line harnesses the Ornstein-Uhlenbeck (OU) mean reversion framework and integrates Natural Language Processing (NLP) sentiment analysis & Large Language Model (LLM) based market shock diagnostics. It consistently outperforms traditional Z-score approaches, particularly in volatile and crisis-prone markets.

However, unlike more developed markets, shorting in Asia can be more difficult, with some countries that have historically imposed short-selling bans or limits. Transaction costs are now also included in this pair trading simulation to assess the model's effectiveness under more realistic market conditions, noting that Asia has above average more pair trades made than the other regions.

In APAC Quant - A Deeper Look at Pair Trading we focus on presenting an application of pair trading as an overlay to long-only portfolios to address both of these concerns. This overlay starts with a long-only baseline. For simplicity, we start by maintaining a long position across all eligible stocks, with capital allocated equally and rebalanced monthly. For stocks not included in the pairs universe, the strategy simply holds a long position. For stocks within the yearly predefined pair list, the OU-based pair trading signal is used to dynamically reallocate capital between the two stocks in each pair. Specifically, where a mean reversion signal would traditionally trigger a short on one leg and a long on the other, the overlay instead shifts capital away from the "would-be shorted" stock and reallocates it to the "would-be long" stock, thus remaining fully invested and compliant within a long-only context.

Figure 39: Pair Overlay Framework



Source: J.P. Morgan Quantitative Strategy

The following charts and tables summarize the empirical results.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Table 11: Long Index with Pair Overlay for MSCI ASIA ex Japan

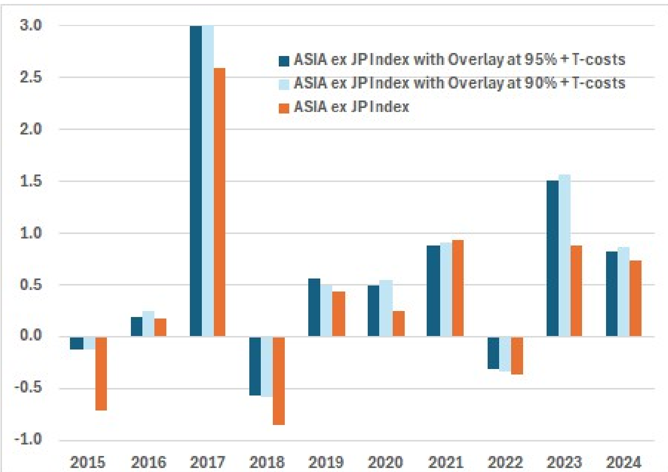
For 2015-2024 **incorporating transaction costs of 20 basis points**

Model	Portfolio Sharpe	Total % P&L	CAGR	Max Drawdown	Avg Vol	Avg Daily P&L Skew	Avg Neg Skew	Avg Pos Skew
ASIA ex JP Index	0.24	30.20	2.67%	-40.48%	9.98%	0.71	-0.70	0.84
Index + Pair Overlay (95%)	0.45	80.86	6.11%	-39.41%	10.79%	0.65	-0.70	0.20
Index + Pair Overlay (90%)	0.46	84.95	6.34%	-39.77%	10.75%	0.68	-0.72	0.31

Source: Source: J.P. Morgan Quantitative Strategy, Factset, MSCI

Figure 40: Long Index Results for MSCI ASIA ex Japan

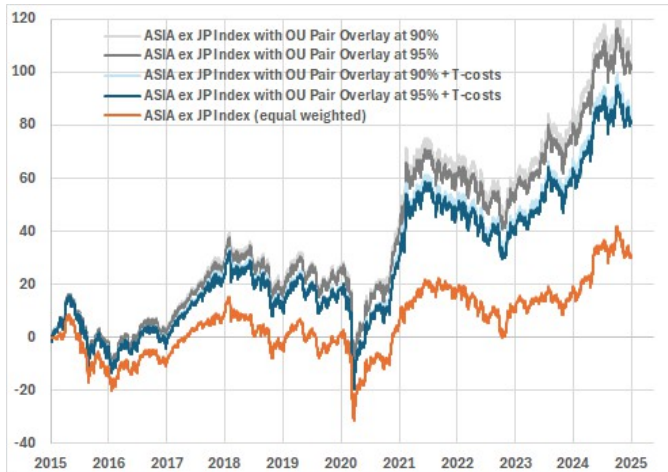
Portfolio Sharpe Ratios for 2015-2024



Source: J.P. Morgan Quantitative Strategy, Factset, MSCI

Figure 41: Portfolio Cumulative P&L for MSCI ASIA ex Japan

Cumulative P&L from 2015-2024, shown as absolute % on starting capital



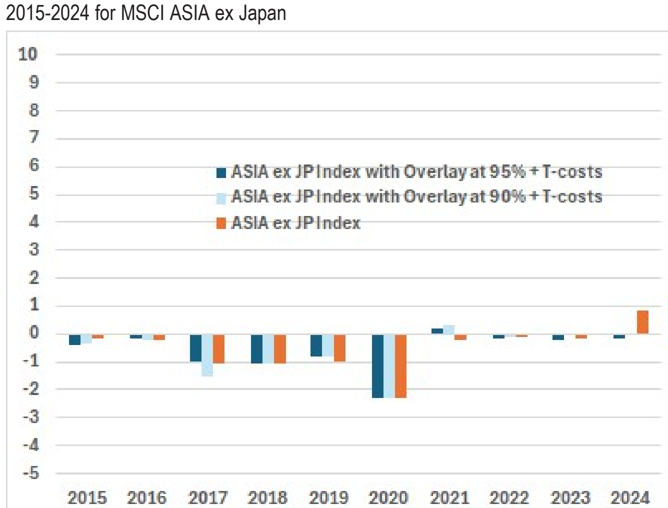
Source: J.P. Morgan Quantitative Strategy, Factset, MSCI

Figure 42: Excess P&L for Pair Overlay

2015-2024 for MSCI ASIA ex Japan

Figure 43: Annual P&L Skewness

2015-2024 for MSCI ASIA ex Japan



Source: J.P. Morgan Quantitative Strategy, Factset, MSCI

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Enhanced trend-following overlay to equity quality

The allocation decision within the equity quality factor

In Enhanced trend overlay to equity quality we revisit the quality factor, prompted by its recent underperformance. It is known that factors are prone to extended periods of underperformance. Is there an alternative to just suffering through such underperformance periods? We recognize the time varying efficacy of styles and seek to vary risk allocated to them, ie the quality sub-factors in this case, on an ex ante basis; style timing. We believe trend following can help because it provides a systematic way to allocate and focus on this avenue of research.

In what follows we take the 4 quality sub-factors used to create the pure quality factor[2] and time them using our trend following signal (discussed here). The rationale is that there is information in the sub-factor allocation decision and a static allocation decision which simply equal weights the scores (as per current quality score build) leaves that information untapped. We also recognise that factors that are established in literature do not go out of fashion quickly. For this reason we take the deliberate decision to not allow shorting in the timed quality sub-factors.

Quality sub-factor FMPs

First, we build the factor mimicking portfolios (FMPs) of the 4 quality sub-factors, RoE, Net Margins, Leverage and Accruals. We include the familiar composite quality factor for reference. By construction, these portfolios have a unit exposure to the targeted factor subject to constraints on market beta, regions, GICS1 sectors and factors [3].

In terms of factors to constrain exposure to, we include the commonly used set of top-level factors, i.e. value, momentum, size and low vol, but instead of constraining the composite quality factor we constrain the 3 quality subfactors not being targeted in each respective build. For example, the RoE portfolio will constrain exposure to the accruals, net margins and leverage factors instead of constraining the composite quality factor.

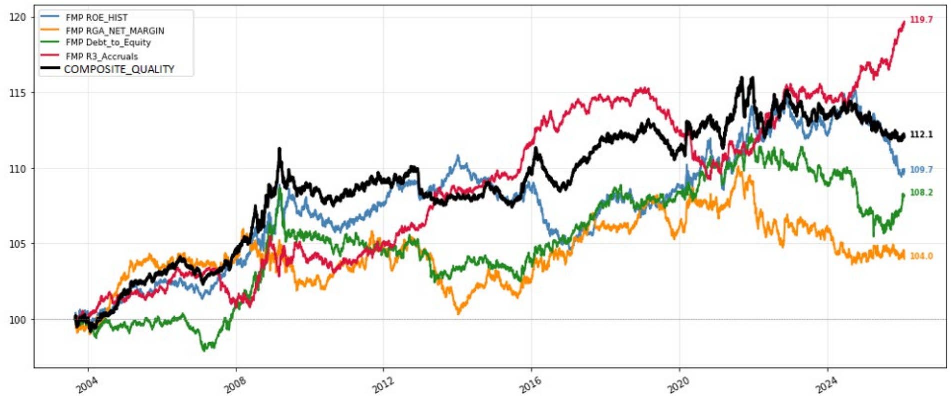
A factor model is used for risk calculations. This includes regions, sectors and all the commonly used factors with the exception of quality. Quality is replaced by its 4 sub-factors. Unlike the by-construction assumption of zero correlation between the common factors, which are pure, we allow a non-zero correlation between the quality sub-factors.

We find that quality has been weak post-Covid. All but one of the quality sub-factors, accruals, are behind this weakness.

2. These are Accruals, Net Margin, ROE and Leverage.
3. More on building pure factor portfolios here

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 44: Compounded Gross Performance of the Pure Factor Mimicking Portfolios



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 12: Gross Performance Statistics for the Pure Factor Mimicking Portfolios

	Composite Quality	Net Margins Sub-factor	Debt to equity Sub-factor	Accruals Sub-factor	RoE Sub-factor
Annualised Return	**0.49%**	0.17%	0.34%	0.78%	0.40%
Annualized Volatility	**1.24%**	1.51%	1.46%	1.16%	1.48%
IR	**0.4**	0.11	0.23	0.67	0.27
Maximum Drawdown	**-3.65%**	-5.93%	-6.06%	-5.30%	-5.64%

Source: J.P. Morgan Cross Asset Systematic Strategy

The trend overlay

The Trend Following Signal is built as discussed in the original Trend-Following note. It is used to either tilt quality sub-factor portfolios directly or tilt bottom-up created exposures. Three different mapping mechanisms are shown in the next chart. Shorting of styles is allowed in some live generic equity factors products. However, here we make a deliberate investment decision not to short any quality sub-factors. We allow zero weights if the signal is extreme enough.

Timing the Quality components

We combine the quality subfactors in two different ways. First, by timing the individually built, pure sub-factor portfolios. One may view an Equal Risk Contribution build as a benchmark here. Based on each of the 4 multipliers, we deviate from an ERC by allocating more/less risk to each of the 4 components, before finally levering up to 5% forecast volatility for the risk budgeted portfolio. Second, we build a bottom-up portfolio that maximizes the volatility *-and trend multiplier-* adjusted exposure to the 4 subfactors while also ensuring that there is multiplier-adjusted risk parity between any pair of sub-factors; the portfolio is, otherwise, pure, so it has no exposure to value, momentum, size, low vol, regions or sectors.

The two mechanisms that are the most long when the signal is positive, mechanisms 2 and 3, do better than the base case in both cases. This supports using the Trend Following Signal as a hedging mechanism which acts on one side only by reducing, either gradually or aggressively, style exposure once the signal turns bearish.

Method 1: Combining portfolios

The composite quality factor portfolio is included for reference. The benchmark ERC build is expected to have volatility of 5% given that this is the volatility (targeted) in the individual sub-factor portfolios. The tilted portfolios, instead of allocating risk equally, scale the risk allocations to the 4 quality blocks up or down based on the multiplier.

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

The simple ERC does better than the composite quality factor. This hints at the composite factor suffering from poor risk allocation between the factors, which is unsurprising given that the composite's aggregate score is merely a sum of z-scores without regard for the relative risk of the 4 subfactors whose scores are getting summed.

Figure 45: Performance of the Risk Budgeted Timed Quality Portfolios - All Tilting Mechanisms

Source: J.P. Morgan Cross Asset Systematic Strategy

Table 13: Performance Statistics for the Different Quality Sub-factor Portfolio Tilting Mechanisms

	Composite Quality	ERC	Asymmetric (1)	Long-only (2)	Linear Long-only (3)
Annualised Return	**0.19%**	**1.42%**	2.18%	2.03%	1.30%
Annualized Volatility	**5.31%**	**5.50%**	5.20%	5.12%	5.11%
IR	**0.04**	**0.26**	0.42	0.4	0.25
Maximum Drawdown	**-26.90%**	**-16.55%**	-15.39%	-17.84%	-18.72%

Source: J.P. Morgan Cross Asset Systematic Strategy

Method 2: A bottom-up build

We modify somewhat the familiar pure portfolio construction methodology to treat the quality factor as a multi-factor comprising of its 4 sub-factors. Furthermore, we take the trend signal multipliers for each FMP into account. We do this in two places. Firstly, we modify the objective function so that the volatility-adjusted exposure of each quality sub-factor is scaled by its respective multiplier. Secondly, we introduce a multiplier-adjusted risk budgeting constraint that ensures the sub-factor exposures are apportioned with respect to the trend-following multipliers.

Results for the different tilting mechanisms are shown below. For this set of tests, the most appropriate benchmark is the composite quality portfolio, which we include as reference. Similarly to what we saw in the quality sub-factor portfolio timing section above, the Asymmetric and Long-only Tilting mechanisms do best.

Table 14: Performance Statistics for the Different Quality Sub-factor Exposures Tilting mechanisms

	Composite Quality	Asymmetric multiplier (1)			Long-only multiplier (2)			Linear Long-only (3)		
Maximum Position Size	50bps	100bps	200bps	300bps	100bps	200bps	300bps	100bps	200bps	300bps
Annualised Return	**0.55%**	1.81%	2.08%	2.05%	1.00%	1.51%	1.66%	-0.03%	0.57%	0.70%
Annualized Volatility	**5.32%**	5.07%	4.91%	4.90%	5.17%	4.96%	4.96%	5.32%	4.98%	5.05%
IR	**0.1**	0.36	0.42	0.42	0.19	0.3	0.34	-0.01	0.11	0.14
Maximum Drawdown	**-26.90%**	-16.62%	-14.85%	-15.71%	-18.91%	-16.57%	-16.87%	-22.91%	-19.00%	-21.85%

Source: J.P. Morgan Cross Asset Systematic Strategy

Profiting from the Third Moment in Single Stocks

Dobromir Tzotchev, PhD
dobromir.tzotchev@jpmorgan.com
J.P. Morgan Securities plc

Mengjiao Wang
mengjiao.wang@jpmorgan.com
J.P. Morgan Securities plc

Rationale and methodology

In Profiting from the third moment in single stocks, we revisit the study of Skewness as factor in single stock space, focusing on the separate analysis of market and residual skewness using a revised methodology. As discussed in the Appendix of Optimal combined skewness signal strategy, the portfolio skewness can be split into the contributions from the third moments of the principal components determined dynamically via empirical Kaiser criterion, and an (uncorrelated) idiosyncratic component. More concretely:

The contribution to portfolio skewness from the j^{th} principal component is

$$skew_{pc_j} = \frac{\left(\sum_i w_i \beta_{ij}\right)^3 M_j}{\sigma_p^3}$$

where $M_j = \frac{1}{T}\sum_t \tilde{r}^3_{pc_j,t}$ is constant, denoting the third moment of the j^{th} principal component, $\sum_i w_i \beta_{ij}$ is the aggregated portfolio beta and σ_p is the portfolio volatility.

We follow the same approach described here and here to construct a portfolio that has the minimum (most negative) skewness contribution from each of the selected principal components individually. That implies that if a principal component exhibits negative skewness, we maximize the portfolio's beta exposure to that component; if the skewness is positive, we minimize the beta exposure. The market skewness portfolio is then constructed as a weighted combination of the portfolios derived from each principal component, where the weight assigned to each component portfolio is proportional to the magnitude of the negative skewness achieved.

The contribution to portfolio skewness from the idiosyncratic part is

$$skew_\epsilon = \frac{\sum_i w_i^3 E_i}{\sigma_p^3}$$

where $E_i = \frac{1}{T}\sum_t \epsilon^3_{i,t}$ is the third moment of the residual returns of asset i.

Minimizing the idiosyncratic skewness components involves cubic powers of the portfolio weights, resulting in a non-convex optimization problem. In our earlier work, we addressed this by decomposing the cubic function into a lower semi-cubic and an upper semi-cubic part. The non-convex lower semi-cubic portion was then approximated in a piecewise linear fashion, allowing us to solve the problem using mixed-integer programming. More details on this approach can be found in the Appendix of Residual skewness as a signal.

However, applying mixed-integer programming to the single stock universe is computationally intensive and becomes infeasible in practice. In this study, we adopt a simplified approach: we minimize a linear combination of the cubic roots of individual asset residual skews, subject to the usual constraints on target volatility and beta-neutrality.

Finally, the optimal combined skewness strategy is constructed as a weighted combination of the skewness exposures achieved through the principal components and the idiosyncratic (residual) part. The weight assigned to each position (whether linked to

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

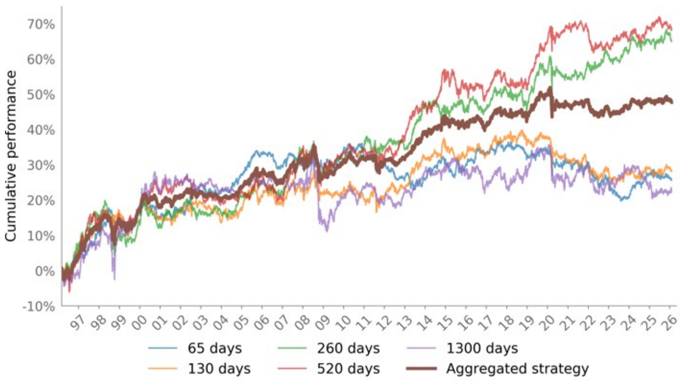
market skewness or residual skewness) is proportional to the magnitude of the negative skewness achieved by that component.

Empirical results

Our investment universe consists of the S&P 500, with trading conducted on a monthly basis. Consistent with the approach in Catching the (third) moment , we focus on 3 months, 6 months, 1 year, 2 years and 5 years lookbacks. Note that the PCA analysis is performed separately for each lookback period and the final positions are the average among the various lookback windows.

Below we present the performance across the various lookback windows as well as the performance of the aggregated strategy.

Figure 46: Cumulative performance of the market skewness strategy applied to single stocks by lookback periods



Source: J.P. Morgan Cross Asset Systematic Strategy

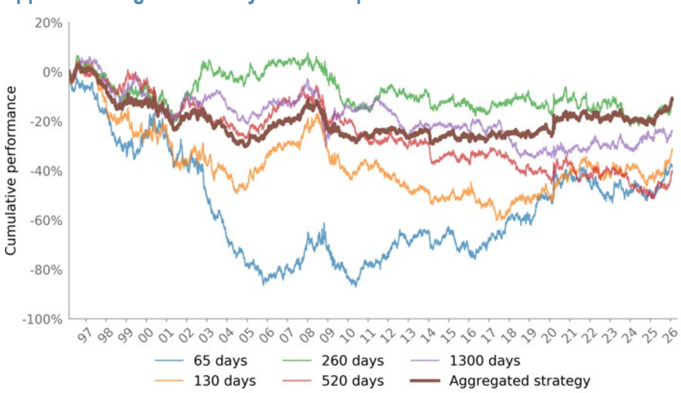
Table 15: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
65 days	0.86%	3.85%	0.22	-16.17%
130 days	0.95%	4.26%	0.22	-14.66%
260 days	2.18%	4.66%	0.47	-14.56%
520 days	2.30%	4.93%	0.47	-11.56%
1300 days	0.74%	5.23%	0.14	-21.97%
Aggregated strategy	1.60%	3.49%	0.46	-10.02%

Source: J.P. Morgan Cross Asset Systematic Strategy

The aggregated market skewness strategy has overall been profitable (Sharpe of 0.46). There appears to be strong diversification benefits among the different lookback periods used for the skewness calculation, with both the very short and the very long-term periods demonstrating inferior performance.

Figure 47: Cumulative performance of the residual skewness strategy applied to single stocks by lookback periods



Source: J.P. Morgan Cross Asset Systematic Strategy

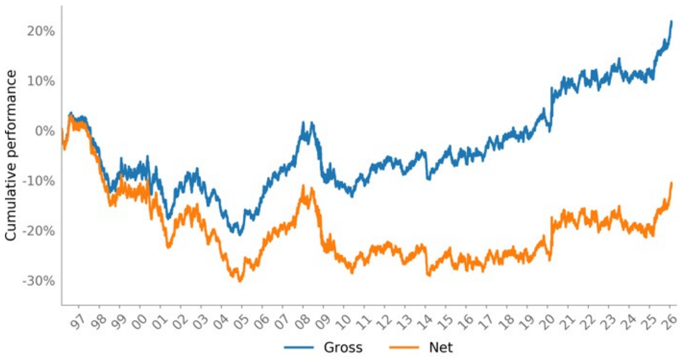
Table 16: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
65 days	-1.29%	7.19%	-0.18	-59.79%
130 days	-1.07%	6.76%	-0.16	-50.40%
260 days	-0.42%	6.17%	-0.07	-28.98%
520 days	-1.35%	5.87%	-0.23	-45.51%
1300 days	-0.80%	5.41%	-0.15	-35.95%
Aggregated strategy	-0.37%	4.42%	-0.08	-29.01%

Source: J.P. Morgan Cross Asset Systematic Strategy

In contrast, the performance of the aggregated residual skewness strategy is overall slightly negative after costs. The correlation between the market and residual skewness approaches is just 0.04.

Figure 48: Cumulative gross and net performance of the aggregated residual skewness strategy



Source: J.P. Morgan Cross Asset Systematic Strategy

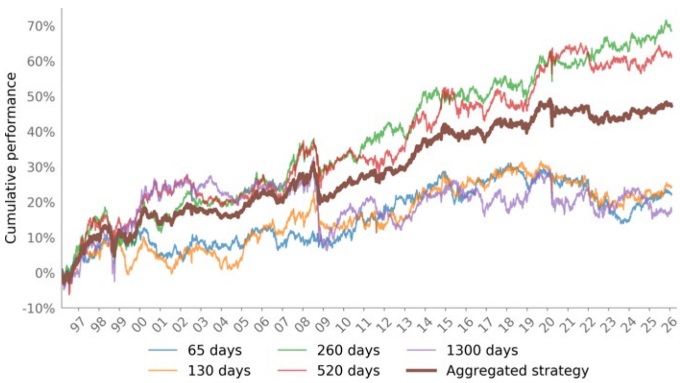
Table 17: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Gross performance	0.72%	4.41%	0.16	-22.51%
Net performance	-0.37%	4.42%	-0.08	-29.01%

Source: J.P. Morgan Cross Asset Systematic Strategy

On a gross basis, the strategy achieves a positive Sharpe ratio of 0.16 over the 20 year backtest, with particularly strong results post GFC. These positive gross returns highlight the return-generating potential of the residual skewness signal. Nevertheless, the strategy's predictive power is insufficient to overcome the impact of significant turnover, which is consistent with our earlier findings in [Skewness as factor in single stock space].

Figure 49: Cumulative performance of the optimal combined skewness strategy applied to single stocks by lookback periods



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 18: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
65 days	0.74%	4.27%	0.17	-16.39%
130 days	0.81%	4.41%	0.18	-13.33%
260 days	2.29%	4.76%	0.48	-14.79%
520 days	2.06%	5.03%	0.41	-13.06%
1300 days	0.58%	5.27%	0.11	-23.69%
Aggregated strategy	1.58%	3.56%	0.44	-11.24%

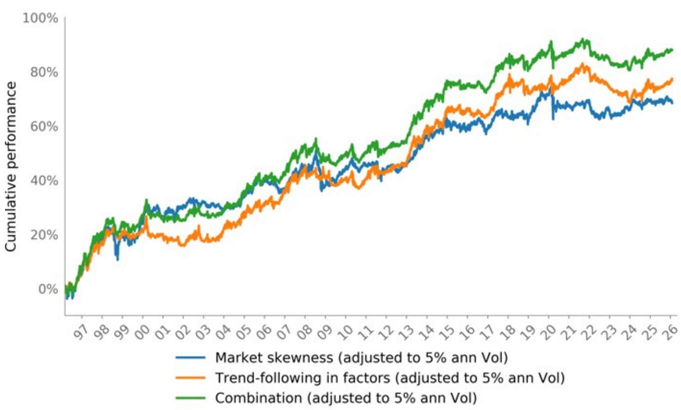
Source: J.P. Morgan Cross Asset Systematic Strategy

The optimized skewness strategy delivers performance comparable to the market skewness strategy, achieving a Sharpe ratio of 0.44. This result further confirms that the market component remains the primary profit driver of the skewness signal approach in the single stock space.

In previous publications we introduced the [trend-following in statistical factors approach applied to single stocks] as well as the [pure single stocks statistical momentum]. As a reminder, the statistical factors are various Principal Components (PCs) extracted from the single stocks universe (with the number of extracted PCs determined by the Empirical Kaiser Criterion). Furthermore, the pure single stocks statistical momentum represents another profit driver that captures the idiosyncratic momentum opportunities (those not reflected in the main PCs).

Below, we present the results of combining market skewness with trend-following in factors, both of which focus on the returns driven by the main statistical factors. The two strategies exhibit a moderately positive correlation of 0.45 over the full sample. While this correlation is somewhat higher than desired, there are still notable diversification benefits from combining the two approaches, as evidenced by a higher Sharpe ratio of 0.59 and a lower drawdown compared to each strategy implemented individually. Please note that a tailored portfolio optimization approach that maximizes market skewness while assuring negative or at least neutral exposure to trend-following in factors can be adopted as well.

Figure 50: Cumulative performance of the market skewness strategy, trend-following in statistical factors and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

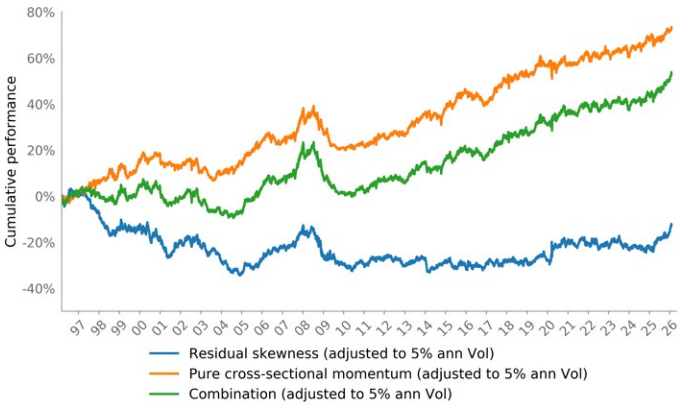
Table 19: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Market skewness (adjusted to 5% ann Vol)	2.29%	5.00%	0.46	-14.13%
Trend-following in factors (adjusted to 5% ann Vol)	2.58%	5.00%	0.52	-13.69%
Combination (adjusted to 5% ann Vol)	2.94%	5.00%	0.59	-11.16%

Source: J.P. Morgan Cross Asset Systematic Strategy

Next, we show the results of combining residual skewness with pure cross-sectional momentum, both of which focus on idiosyncratic returns. The two strategies exhibit a small positive correlation of 0.16; however, the flattish performance of the residual skewness outweights the diversification benefits in the combination.

Figure 51: Cumulative performance of the residual skewness strategy, pure cross-sectional momentum and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

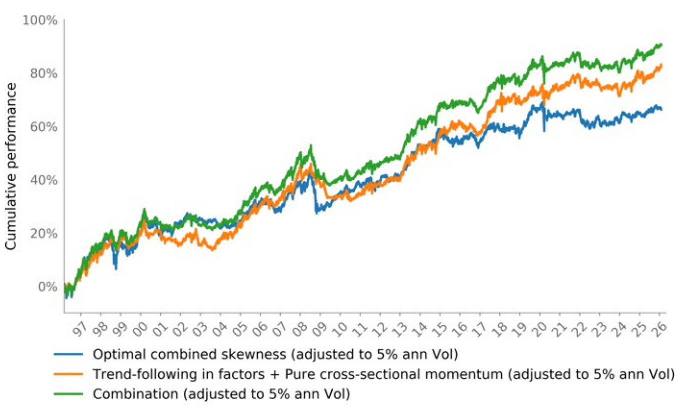
Table 20: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Residual skewness (adjusted to 5% ann Vol)	-0.42%	5.00%	-0.08	-32.25%
Pure cross-sectional momentum (adjusted to 5% ann Vol)	2.44%	5.00%	0.49	-17.71%
Combination (adjusted to 5% ann Vol)	1.78%	5.00%	0.36	-21.36%

Source: J.P. Morgan Cross Asset Systematic Strategy

Finally, we combine the optimal combined skewness strategy with a portfolio consisting of trend-following in factors and pure cross-sectional momentum. The aggregated portfolio delivers a robust Sharpe ratio of 0.61, which is very similar to the combined strategy of market skewness and trend-following in factors. This result re-confirms that, within the single stock space, investors may benefit most by focusing on profit opportunities driven by the skewness of the market components.

Figure 52: Cumulative performance of the optimal combined skewness strategy, trend-following in statistical factors & pure cross-sectional momentum and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 21: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Optimal combined skewness (adjusted to 5% ann Vol)	2.22%	5.00%	0.44	-15.47%
Trend-following in factors + Pure cross-sectional momentum (adjusted to 5% ann Vol)	2.77%	5.00%	0.55	-13.33%
Combination (adjusted to 5% ann Vol)	3.03%	5.00%	0.61	-14.22%

Source: J.P. Morgan Cross Asset Systematic Strategy

Profiting from the Third Moment in Generic Equity Factors (GEFs)

Dobromir Tzotchev, PhD
dobromir.tzotchev@jpmorgan.com
J.P. Morgan Securities plc

Mengjiao Wang
mengjiao.wang@jpmorgan.com
J.P. Morgan Securities plc

Rationale and methodology

In Trend-following in Generic Equity Factors, we introduced an application of the trend-following framework to a diversified set of Generic Equity Factors (GEFs). These GEFs are constructed to be delta-, sector-, and beta-neutral, and are based on primary factor metrics. Drawing parallels with trend-following in statistical factors and pure statistical cross-sectional momentum applied to single stocks, we showed that trend-following in GEFs is uncorrelated with trend-following in the first principal component, making it an effective diversifier to traditional trend-following strategies in equity markets.

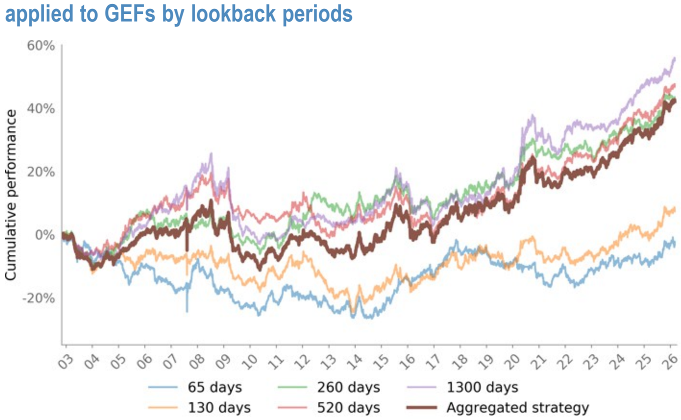
In Profiting from the Third Moment in Generic Equity Factors (GEFs), we extend our skewness-reversion signal framework to GEFs. As a reminder, we distinguish between two profit drivers in our skewness strategy - market skewness and residual skewness. Market skewness component is determined by the third moment of the principal components extracted dynamically via the Empirical Kaiser criterion, while residual skewness is driven by the (uncorrelated) idiosyncratic component. We have also put forward an optimal skewness approach that combines market and residual components based on the profit potential embedded in each of those strategies.

Consistent with the approach in Catching the (third) moment , we focus on 3 months, 6 months, 1 year, 2 years and 5 years lookbacks. Note that the PCA analysis is performed separately for each lookback period and the final positions are the average among the various lookback windows.

Empirical results

Below we present the performance across the various lookback windows as well as the performance of the aggregated strategy.

Figure 53: Cumulative performance of the market skewness strategy applied to GEFs by lookback periods



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 22: Performance statistics

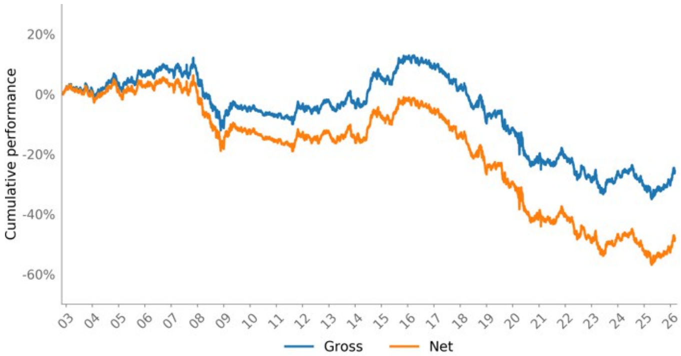
	Ann Return	Ann Volatility	Sharpe	Max DD
65 days	-0.10%	4.46%	-0.02	-24.43%
130 days	0.30%	4.46%	0.07	-23.25%
260 days	1.78%	4.90%	0.36	-13.76%
520 days	1.94%	4.98%	0.39	-19.06%
1300 days	2.28%	5.32%	0.43	-25.43%
Aggregated strategy	1.74%	5.09%	0.34	-20.15%

Source: J.P. Morgan Cross Asset Systematic Strategy

The aggregated market skewness strategy has been profitable overall, delivering a Sharpe ratio of 0.34. Diversification across different lookback windows is strong, with shorter lookbacks tending to underperform relative to longer horizons (as a reminder, in the case of single stocks the performance of the different lookbacks was quite uniform).

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 54: Cumulative gross and net performance of the aggregated residual skewness strategy applied to GEFs



Source: J.P. Morgan Cross Asset Systematic Strategy

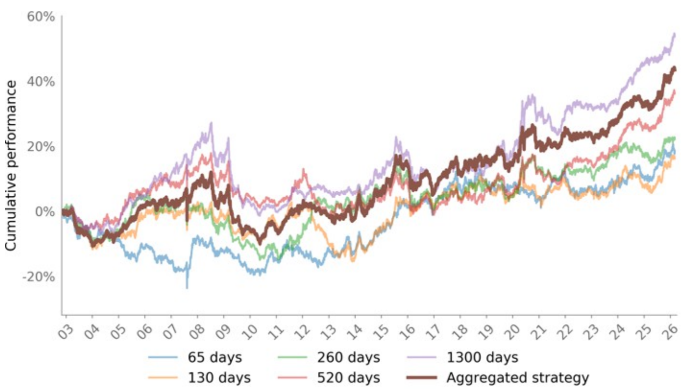
Table 23: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Gross performance	-1.06%	5.14%	-0.21	-39.15%
Net performance	-2.00%	5.14%	-0.39	-48.15%

Source: J.P. Morgan Cross Asset Systematic Strategy

In contrast, the aggregated residual skewness strategy has delivered negative performance, consistent with our earlier findings in single stocks. Notably, in the GEFs context even gross performance is negative, whereas in single stocks the gross Sharpe was moderately positive over a 20-year backtest, with particularly strong results post-GFC.

Figure 55: Cumulative performance of the optimal combined skewness strategy applied to GEFs by lookback periods



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 24: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
65 days	0.77%	4.82%	0.16	-21.52%
130 days	0.67%	4.67%	0.14	-17.29%
260 days	0.91%	4.91%	0.19	-18.83%
520 days	1.49%	4.94%	0.30	-18.32%
1300 days	2.22%	5.33%	0.42	-25.34%
Aggregated strategy	1.80%	5.10%	0.35	-20.08%

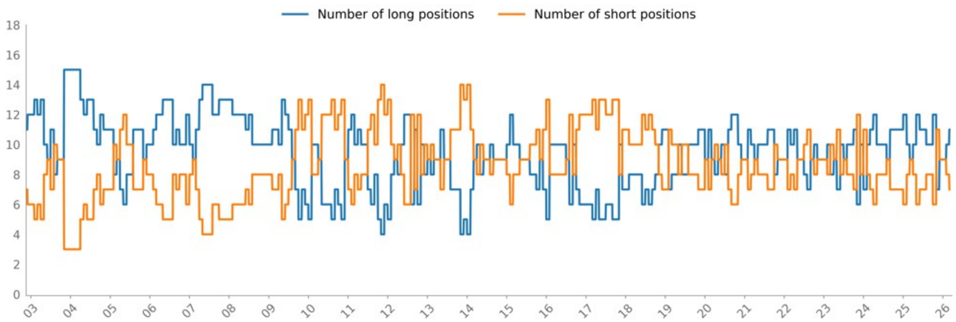
Source: J.P. Morgan Cross Asset Systematic Strategy

The optimized skewness strategy (the one that allocates simultaneously to market skewness and residual skewness portfolios in proportion to the magnitude of the respective skewness signals) delivers performance comparable to the market skewness strategy, with a Sharpe ratio of 0.35. This aligns with our findings in Profiting from the Third Moment in Single Stocks and further corroborates that, in equity markets, the efficacy of the skewness signal is primarily driven by market skewness.

In the graph below, we show the number of long and short equity factor positions over time. The average gap between the longs and the shorts is only about 1 (out of 18 factors).

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 56: Number of long/short GEFs positions

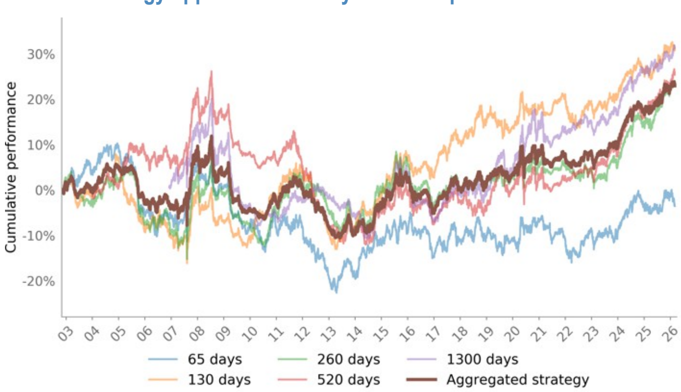


Source: J.P. Morgan Cross Asset Systematic Strategy

Be mindful that, given the structural design of GEFs, common market and sector co-movements are intentionally eliminated, which tends to reduce cross-factor correlations. As a result, the leading principal components may cluster within specific styles rather than revealing broad, market-wide drivers of variance. This structural feature can limit the breadth of the "market" component and should be kept in mind when separately analyzing market and residual skewness.

Accordingly, we also apply the portfolio construction described in Catching the (third) moment, which uses the skewness of GEFs returns as a signal: we take positions proportional to $-1.0 \times$ skew for each factor and then scale the portfolio to a 5% target volatility. We then compare this baseline approach with the optimal combined skewness strategy.

Figure 57: Cumulative performance of the skewness as a signal baseline strategy applied to GEFs by lookback periods



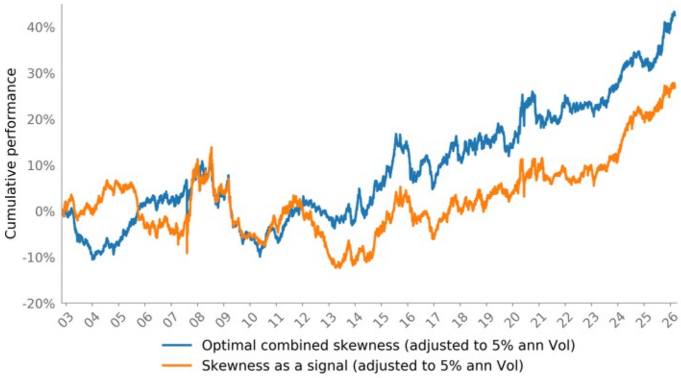
Source: J.P. Morgan Cross Asset Systematic Strategy

Table 25: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
65 days	0.12%	4.63%	0.03	-30.58%
130 days	1.19%	4.69%	0.25	-20.55%
260 days	1.11%	4.97%	0.22	-18.72%
520 days	1.13%	5.04%	0.22	-34.19%
1300 days	1.63%	5.42%	0.30	-25.93%
Aggregated strategy	1.02%	4.22%	0.24	-21.65%

Source: J.P. Morgan Cross Asset Systematic Strategy

Figure 58: Cumulative performance of the optimal combined skewness strategy and the skewness as a signal baseline strategy



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 26: Performance statistics

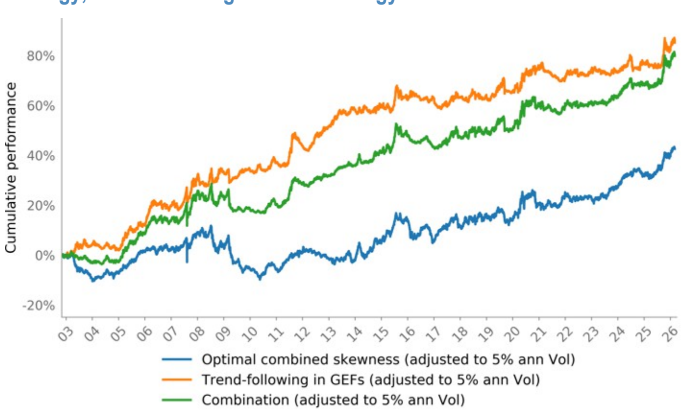
	Ann Return	Ann Volatility	Sharpe	Max DD
Optimal combined skewness (adjusted to 5% ann Vol)	1.76%	5.00%	0.35	-19.72%
Skewness as a signal (adjusted to 5% ann Vol)	1.20%	5.00%	0.24	-25.14%

Source: J.P. Morgan Cross Asset Systematic Strategy

With the skewness as a signal approach, the 65-day lookback exhibits weak results, while the other windows deliver similar Sharpe ratios in the region of 0.2-0.3. The aggregated skewness as a signal strategy is highly correlated with the optimal combined skewness strategy (correlation 0.74) but delivers inferior performance over the same backtest period (Sharpe 0.24 versus 0.35).

Finally, we present the results of combining the optimal combined skewness strategy with trend-following in GEFs. The two strategies exhibit a moderately positive correlation of 0.38 over the full sample. The combined portfolio delivers a Sharpe ratio of 0.66, which is slightly below the standalone performance of trend-following in GEFs. However, the optimal combined skewness strategy shows a more persistently positive momentum than trend-following in GEFs after 2013, resulting in steady gains for the combined approach. On this basis, we remain confident that the skewness strategy will provide strong diversification potential to the trend-following strategy going forward.

Figure 59: Cumulative performance of the optimal combined skewness strategy, trend-following in GEFs strategy and their combination



Source: J.P. Morgan Cross Asset Systematic Strategy

Table 27: Performance statistics

	Ann Return	Ann Volatility	Sharpe	Max DD
Optimal combined skewness (adjusted to 5% ann Vol)	1.76%	5.00%	0.35	-19.72%
Trend-following in GEFs (adjusted to 5% ann Vol)	3.53%	5.00%	0.71	-9.14%
Combination (adjusted to 5% ann Vol)	3.31%	5.00%	0.66	-11.27%

Source: J.P. Morgan Cross Asset Systematic Strategy

Derivatives Quant Research and Systematic Strategies

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Yangyang Hou
yangyang.hou@jpmorgan.com
J.P. Morgan Securities plc

Mengdi Wang
mengdi.wang@jpmorgan.com
J.P. Morgan Securities LLC

What To Long, Short, and When (Part II)

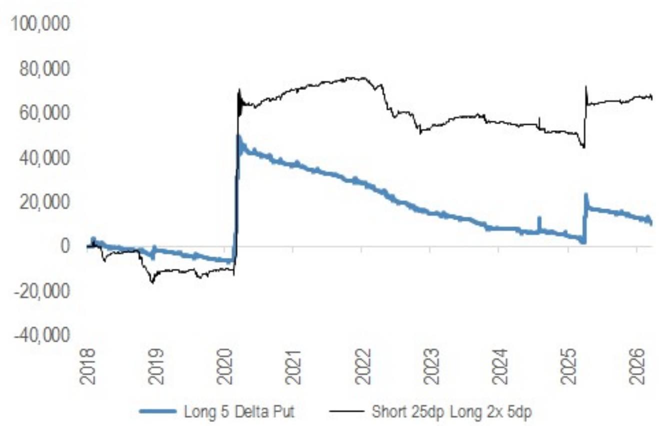
In *What to Long, Short and When (Part II)*, we construct a systematic tail hedging solution that is always net long gamma by the close, with minimal bleed. The framework uses the flip side of the "safer carry" findings from Part I (link) to build a defensive portfolio with long gamma, long convexity and short skew exposure.

Tail Protection And Cost

The 3M 5-delta puts are the ideal convexity leg. In flash crashes (e.g. 2020 and 2025) when spot fell near the strikelevels, volatility realized around 54 vols , far exceeding the implied levels of around 30, and delivering outsized payoffs relative to cost. JPM Quant Strategist identified a suitable funding leg to be 3M 25 delta put theta neutral, approximately a 1x2 put ratio, that delivers in high-sigma crashes. However, the structure suffers during "**slow death**" markets (e.g. late 2018, 2022), where net short gamma dominates before the tail leg kicks in.

Figure 60: 3M Short 25 dp Long 2x 5dp VS Long 5 Delta Put Only

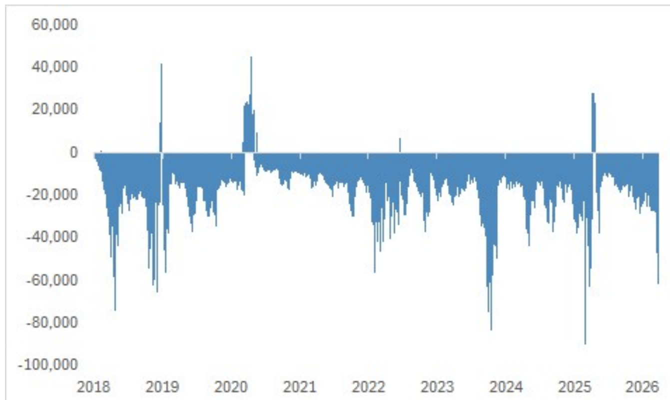
$1M notional daily scaled by tenor



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

Figure 61: Net Gamma Exposure for 3M 25-5 Delta 1x2 PutRatio

$1M notional daily scaled by tenor



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

A Portfolio That Is Always Long Gamma

To protect against mild corrections without excess bleed, we proposed to **own gamma by long a blend of ATM and 25 delta puts funded by short 10 delta**, at a discount by exploiting short skew. This creates continuous long gamma coverage from ~97%-104% of spot. The overlay addresses most of the short gamma issue from the put ratio, with minor drawdowns from short skew covered by the convexity leg.

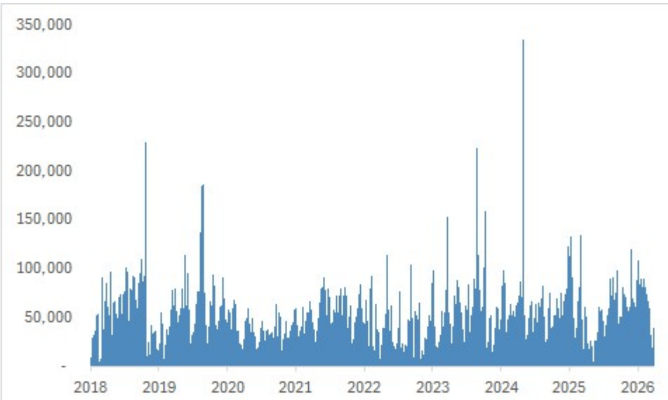
However, short gamma episodes may still occur during distressed markets, which can be captured by term structure inversion. Hence, a "gamma patch" using 2W ATM-25dp put spreads is triggered when the term structure inverts. Additionally, for the few residual days of negative gamma, a 5-day 25delta call is sized to bring portfolio gamma to zero by the close. The final three-leg , construction is as follows, with all delta hedged daily till expiry:

- **Long Convexity:** Daily 3M put ratio of long 2x 5dp, short 1x 25dp to fund, size 1/tenor

- **Long Gamma:** Daily 1M put spread of long 0.5x 50dp, long 0.5x 25dp, short 1x 10dp to fund, size 1/tenor
- **Extra Gamma Patches:** When 1M-3M term structure inverts, long 2W 50-25dp put spread (size 1/tenor). If net gamma is still negative prior to close, buy 5-day 25dc and size to make portfolio gamma 0.

Figure 62: Final Portfolio is Always Long Gamma

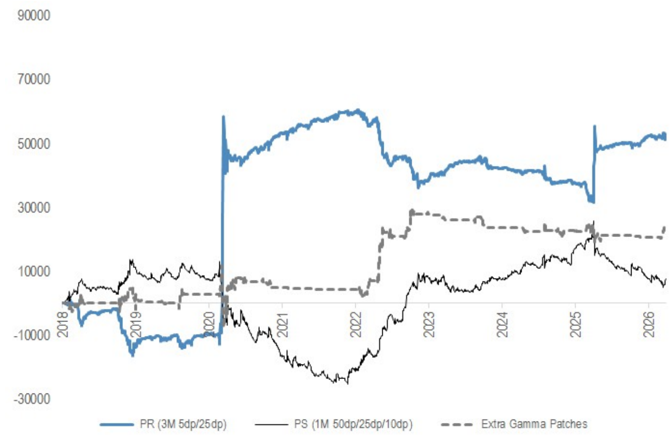
$1M notional daily scaled by tenor



Source: J.P. Morgan Equity Derivatives Strategy, ORPA, Bloomberg Finance L.P

Figure 63: PnL Split of the Three Legs

$1M notional daily scaled by tenor



Source: J.P. Morgan Equity Derivatives Strategy, ORPA, Bloomberg Finance L.P.

The final tail portfolio delivered robust orthogonal returns to SPX in nearly all corrections since 2018 with minimal bleed, achieving a Sharpe ratio of 0.7 and worst DD of -1.8%.

Out-of-Sample Tests

To confirm the framework is not overfitted, we tested on SX5E and NKY over the same period, and on SPX extended back to 2008. Across all indices, the convexity and gamma legs exhibit complementary behavior: the put ratio captures sharp, high-sigma moves (COVID, Yen Carry Unwind, Liberation Day) while the gamma leg performs during grinding declines.

Figure 64: Total "Always Long Gamma" Tail Portfolio Pnl vs SPX

Left - Tail Pnl, Right - SPX Spot

Source: J.P. Morgan Equity Derivatives Strategy, ORPA, Bloomberg Finance L.P.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 65: SX5E Backtest PnL vs Spot

Left - Tail Pnl, Right - SX5E Spot

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 66: NKY Backtest PnL vs Spot

Left - Tail Pnl, Right - NKY Spot

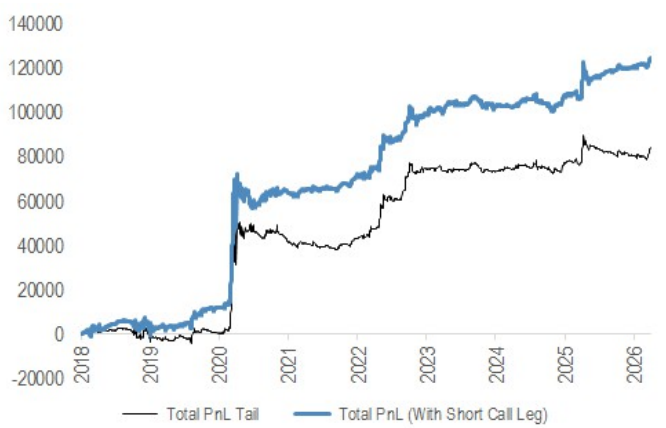
Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Further Enhancements

Being long gamma at all times may be excessive when the risk premium for gamma is too rich. Overlaying a 1M 10 delta call overwriting with enhancements (from Part I findings) funds excess bleed during strong bull markets (2021, 2025). This improves the Sharpe ratio to 0.92 with max DD of -2.06%. As a trade-off, portfolio gamma is no longer always positive, but negative gamma only materializes on the upside during periods when SPX is grinding higher. This remains a comfortable position given the call leg's embedded features to avoid excess short gamma around the strike.

Figure 67: Portfolio with Upside DH Call Overlay

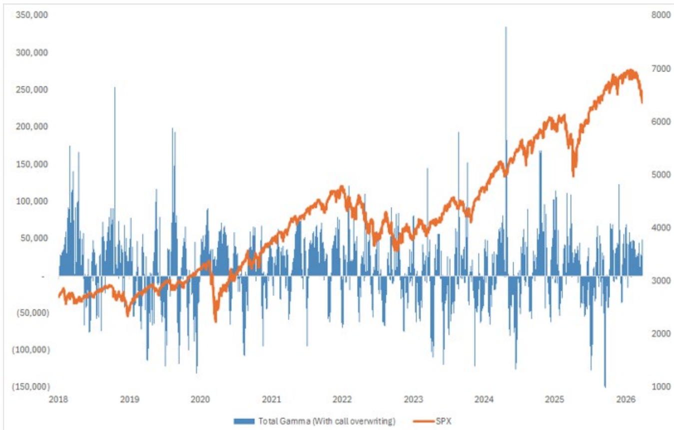
$1M notional daily scaled by tenor



Source: J.P. Morgan Equity Derivatives Strategy, ORPA, Bloomberg Finance L.P.

Figure 68: Portfolio Gamma Including Short Call vs SPX

Left - Gamma, Right - SPX Spot



Source: J.P. Morgan Equity Derivatives Strategy, ORPA, Bloomberg Finance L.P.

Conclusion

Short skew, while seemingly risky, has been one of the best-performing vol trades in 2022 and recent correction, and serves as an effective portfolio funding when paired with long gamma and convexity legs across staggered strikes. Similarly, short upside calls can provide additional alpha when positioned on the right part of the curve with disciplined entry and exit timing. *Note: No transaction costs are included in the backtest results presented.*

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Yangyang Hou
yangyang.hou@jpmorgan.com
J.P. Morgan Securities plc

Mengdi Wang
mengdi.wang@jpmorgan.com
J.P. Morgan Securities LLC

Systematic Option Strategies in VIX

In *Lessons From The 2022 Playbook (link),* we discussed the similarities of the March correction compared to 2022 and flag the best performing vol trade being short skew and short vol of vol. We investigate the latter via options on the VIX and identify systematic opportunities for carry and tail.

As with any risk premium harvesting strategy, the key is to reside in a region where **the probability of large drawdowns can be minimized**. We followed a similar approach to previous **"safer" SPX carry strategy** (link) and ran the exercise for selling delta-hedged option on VIX.

Figure 69: Sharpe Ratio of Delta Hedged VIX Options by Strike and Tenor

Jan-2017 to Aug-25, Unwind 1W Prior to Expiry Due to Liquidity, No Transaction Cost

	10d_P	25d_P	50d_S	25d_C	10d_C
1M	0.48	0.67	1.10	0.87	0.38
2M	0.59	0.65	0.64	0.28	-0.06
3M	0.64	0.62	0.56	0.26	-0.07

Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy, OPRA

In the risk premium space, one can think of the **downside of VIX options as somewhat similar to the upside of SPX options,** where we identified to be the "safer" regions to harvest.

Figure 70: Average Initial Strike Prices by Strike and Tenor

	10d_P	25d_P	50d_S	25d_C	10d_C
1M	15.3	17.5	21.2	28.8	43.1
2M	15.0	17.6	22.4	32.7	52.1
3M	14.8	17.6	23.0	34.4	55.9

Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy, OPRA

Figure 71: Hit Ratio of Delta Hedged VIX Options by Strike and Tenor

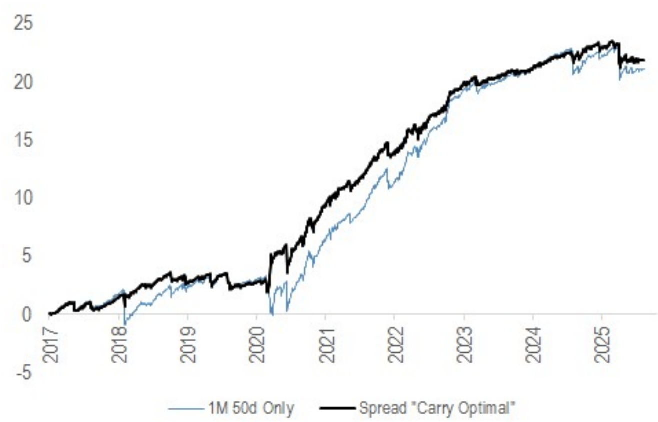
	10d_P	25d_P	50d_S	25d_C	10d_C
1M	61%	62%	78%	85%	89%
2M	64%	69%	76%	80%	82%
3M	67%	68%	74%	80%	81%

Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy, OPRA

Therefore, we find that the **best Sharpe ratios come from the front months ATM to downside strikes on VIX** (Figure 69) despite having lower hit ratios than the upside wings, where large drawdowns wipe out the high hit ratios from the carry (Figure 71).

Figure 72: Short 1M 50d Straddle Long 1x 3M 10dC on VIX Option

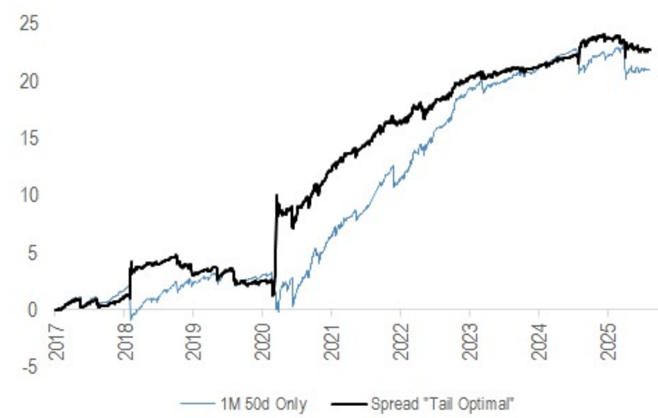
Delta hedge daily unwind both legs 1W prior expiry



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy, OPRA

Figure 73: Short 1M 50d Straddle Long 2x 3M 10dC on VIX Option

Delta hedge daily unwind both legs 1W prior expiry



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy, OPRA

We can make use of the 3M 10dC as a **tail hedge** and build a more conservative systematic vol of vol carry. Using 1M 50dC as the base strategy, for PMs with **a carry-focused mandate,** 1x participation in the tail hedge has been sufficient to reduce drawdowns (Figure 72).

For those who **focus on tail mandates, at least 2x** on the tail participation would make the strategy close to day 1 gamma neutral, and generate a tail-like profile without excessive bleed (Figure 73).

Figure 74: Sharpe Ratio and Worst Drawdown for Vol of Vol Carry and with Tail Hedge

	Short 1M 50dS	Short 1M 50dS	Short 1M 50dS
		Long 1x 3M 10dC	Long 2x 3M 10dC
Sharpe	1.1	1.4	1.15
Worst DD	-3.65	-2.3	-3.49

Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy, OPRA

By including the tail leg, the carry strategy achieved higher Sharpe ratios and lower worst drawdowns.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Yangyang Hou
yangyang.hou@jpmorgan.com
J.P. Morgan Securities plc

Ladislav Jankovic
ladislav.jankovic@jpmorgan.com
J.P. Morgan Securities LLC

Mengdi Wang
mengdi.wang@jpmorgan.com
J.P. Morgan Securities LLC

Systematic Trades in Oil and Gold Options

When the war first broke out, we looked at the dislocations in the oil option market and identified good entry level for a calendar position in Brent for both tactical and systematic implementations (link). In Gold options, a similar short ATM long wing position has been profitable (link).

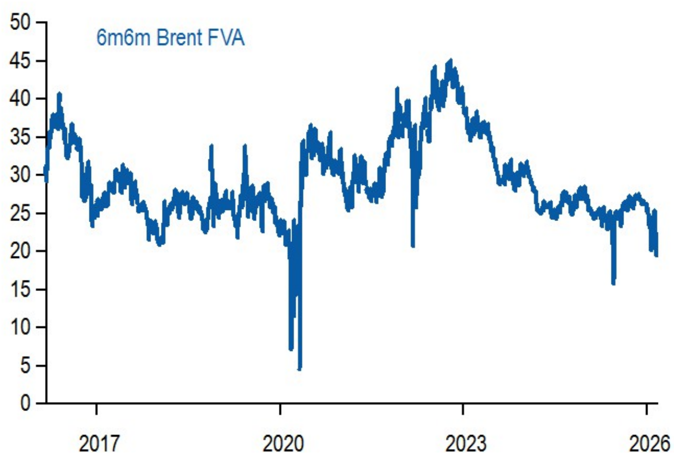
Brent -M6 / +M12 calendar

(As of Mar 2026) The pickup in vol curve inversion has pushed fwd vols below 20—levels only seen during the COVID shock and briefly in 2025 (Figure 75) . The dislocation between front and back vols implies that vol curve RV trades offer attractive entry points, especially when structured **gamma neutral** to avoid exposure to short-term geopolitical risk. Skew is sharply higher as well, but the firm pricing is concentrated in the front, leaving back-end skew relatively benignly priced (Figure 76). This allows for owning back topside vols at more reasonable prices, further enhancing the risk-reward profile of the trade.

Given these dynamics, a short M6 / long M12 Brent calendar spread, structured gamma neutral and long vega, is well supported. The trade benefits from the normalization of front end vols, and effectively hedging that via a back end skew long exposure. Historical performance of similar constructs has been robust (Figure 77). Implementing the trade as a buy of M12 25 delta call versus a sell of M6 ATM call in **gamma neutral notionals** captures these relative value opportunities while maintaining risks in check.

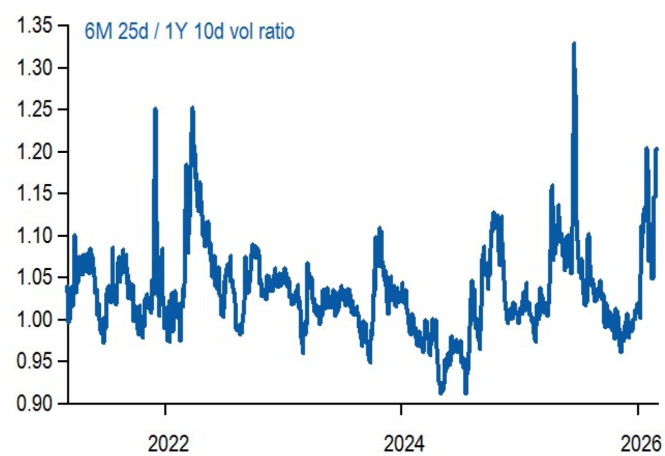
— Buy delta-hedged M12 25 delta call @29% indic vs sell M6 atm call @36.6%, in gamma neutral notionals

Figure 75: Fwd vols dropped to below 20 vols.



Source: J.P. Morgan Global Markets Strategy.

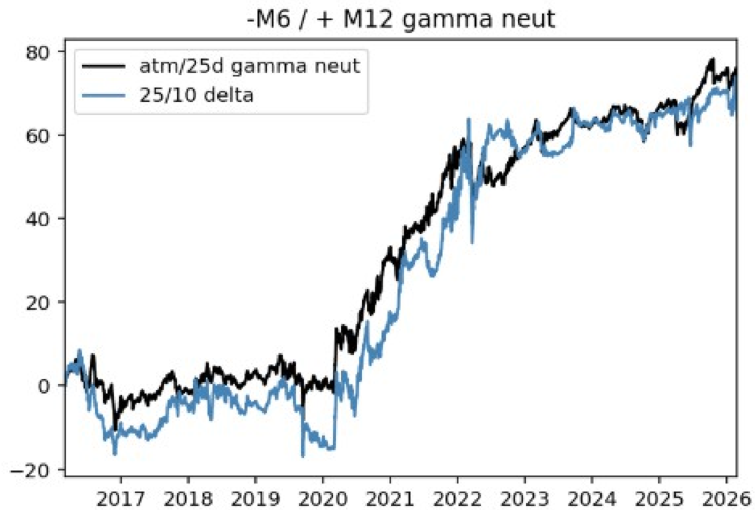
Figure 76: Skew popped higher, but largely so in the front.



Source: J.P. Morgan Global Markets Strategy.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Figure 77: Cummulative returns from -M6 / +M12 gamma neutral vega long calendar constructs that are also long skew have been holding well.



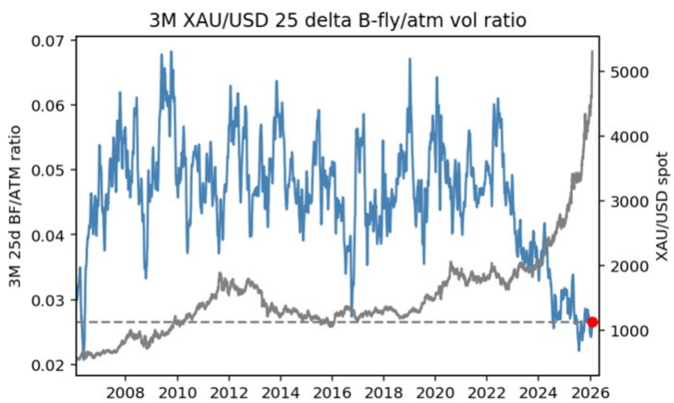
Source: J.P. Morgan Global Markets Strategy.

Gold Vol Surface RV Trading

(As of Jan 2026) Gold has rallied nearly 100% over the past year, briefly breaking through the 5500 level this week before today's spectacular selloff. While XAU vols have remained relatively contained in 2025 except for a few spikes, the latest leg of price action broke its back and has seen vol curve sharply reprice higher (1M vol above 35—placing it in the high-vol tails of the 5-year distribution). Interestingly, the fly/atm vol ratio is near a 20-year low (Figure 78) which motivates selling delta-hedged XAU/USD straddles and being long strangles. These types of structures have been effective for reducing downside risk while capturing convexity and spot-vol corr over the last two years.

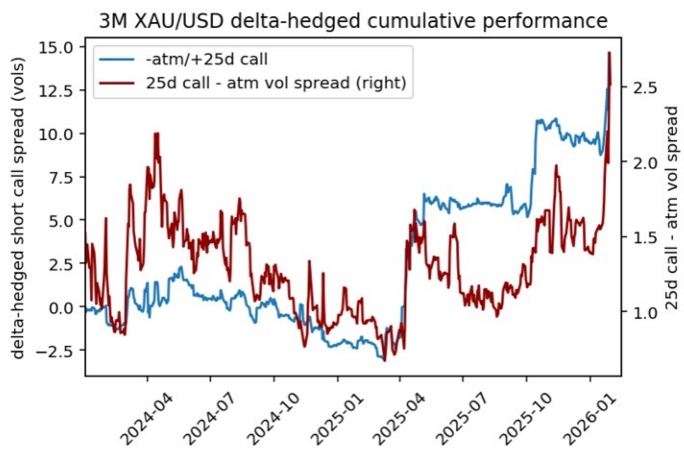
The backtest (Figure 79) shows potive returns during the 2024-2026 period thanks to long otm call leg offseting short gamma downside, and leveraging skew - atm repricing as well as via realized spot-vol corr channel. Once the dust settles after today's spot moves, we believe the setup will be favorable for once again ***selling delta-hedged 2M atm XAU/USD call @32.25/34.25indic vs buy 6M 25 delta call @30ch***, equal notionals. The choice of tenors is attempting to maximize the effectiveness of the hedge (6M leg) vis-à-vis short atm at 2M for when the spot dynamic eases off.

Figure 78: Despite the unprecedented rally in gold spot price, flies / atm vol ratio is near a 20 year low



Source: J.P. Morgan.

Figure 79: Short atm / long convexity delta-hedged XAU/USD structures have been efficient in capturing spot-vol dynamics in the last two years.



Source: J.P. Morgan.

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

List of 2026 reports with focus on systematic investing

Team	Lead AC	Title	Subtitle
Global FX Strategy	Antonin T Delair	Rates Macro Quant: Forecast Revisions as a Measure of Macro Momentum	Forecast Revisions as a Measure of Macro Momentum
Global Quantitative & Derivatives Strategy	Yangyang Hou	What to Long, Short, and When (Part II)	What to Long, Short, and When (Part II)
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	ARPs during geo-political stress	Trend overlay to equity quality, ARPs during geo-political stress, Profiting from the third moment in GEFs
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Enhanced trend overlay to equity quality	Trend overlay to equity quality, ARPs during geo-political stress, Profiting from the third moment in GEFs
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Profiting from the third moment in GEFs	Trend overlay to equity quality, ARPs during geo-political stress, Profiting from the third moment in GEFs
Global Quantitative & Derivatives Strategy	Yangyang Hou	Systematic Options Strategies in VIX	Lesson from the 2022 Playbookk
Global Quantitative & Derivatives Strategy	Yangyang Hou	Systematic Trades in Oil and Gold Options	Oil Derivatives and “Cheaper To Deliver” Macro Hedges
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Profiting from the third moment in single stocks	Profiting from the third moment in single stocks and latest timing recommendations - OW Rates&Equity Value, and Credit Carry&Short Volatility
Global Cross Asset Strategy	Thomas Salopek	Deep Momentum	Deep Momentum Models
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Deep dive into the 2025 performance of the JPM Research risk premia strategies	2025 review, QIS implementation of the 2026 cross-asset views, Optimized trend-following in factors
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	QIS implementation of the 2026 cross-asset views	2025 review, QIS implementation of the 2026 cross-asset views, Optimized trend-following in factors
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Optimized trend-following in factors	2025 review, QIS implementation of the 2026 cross-asset views, Optimized trend-following in factors
Global Quantitative Strategy	Robert Smith, PhD	APAC Quant - A Deeper Look at Pair Trading	Implementing Pairs in Asia (ex Japan) with Market Constraints

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

List of 2025 reports with focus on systematic investing

Team	Lead AC	Title	Subtitle
Global Quantitative Strategy	Robert Smith, PhD	APAC Quant - A Deeper Look at Pair Trading	Implementing Pairs in Asia (ex Japan) with Market Constraints
Global FX Strategy	Antonin T Delair	Rates Macro Quant	Swaps vs Bond Futures: the carry use-case
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Systematic Highlights	Catch some waves… Combining our Regime model with momentum in QIS and public markets
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Implications of the 2026 outlooks for QIS investors and pure cross-asset momentum within&across asset classes
Global Quantitative Strategy	Robert Smith, PhD	BuzzMiner for Global News	Major themes for markets from tweets (week ending 30 November)
Global Quantitative Strategy	Robert Smith, PhD	BuzzMiner for Global News (X edition)	Major Themes for Markets from Tweets (Week ending 09 November)
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Revisiting trend-following in cross-asset factors and crowding as a timing signal for the Risk Parity portfolio
Emerging Markets Strategy	Tales Padilha	EM Quant Series	Introducing our new framework for assessing long-term fair value in REERs
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Systematic Highlights	Less Is More: Factor Investing by PCA and Its Variants
European Equity Quantitative Strategy	William Summer	European Quant Strategy	Human Capital Meets Multi-Factor Model
Global Quantitative & Derivatives Strategy	Yangyang Hou	Equity Volatility Strategy	Dispersion Update, AI vs Dot-Com and Earnings
Global FX Strategy	Antonin T Delair	Rates Macro Quant	Rates ‘real’ carry with a duration overlay
Global FX Strategy	Antonin T Delair	Rates Macro Quant	Swaps vs Bond Futures: the carry use-case
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Systematic Highlights	Breadth of Crowding Metrics
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Systematic Highlights	Improving our TAA: a more responsive Risk Premia by Pairs
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Incorporating off-benchmark commodities into our skewness strategy and quantifying crowding in trend-following strategies
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Trend-Following in Generic Equity Factors
Global Quantitative Strategy	Robert Smith, PhD	BuzzMiner for Global News (X edition)	Major Themes for Markets from Tweets (Week ending 10 August)
Global Quantitative & Derivatives Strategy	Juan Duran-Vara	FX Derivatives	EM vol calendars - portfolio characteristics

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Emma Wu	From Vols to Correlation	A Forward-Looking Framework for FX Correlation Trading and Cross Asset Portfolio Integration
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspective on Cross-Asset Risk Premia	Incorporating momentum, positioning and fair value signals in carry strategies
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Trend-following synopsis and timing equity factors with combined macro and trend signals
Global Quantitative & Derivatives Strategy	Emma Wu	Equity Volatility Strategy	Thinking Outside the Box for Tail Trading
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Systematic Highlights	Cross Asset Momentum Spillover
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspective on Cross-Asset Risk Premia	Focus on trend-following in off-benchmark commodities, and latest timing model update - OW Rates Value and Momentum, and UW Equity Short Vol
Emerging Markets Strategy	Tales Padilha	EM Quant Series	Monitoring Fair Value in Sovereign Credit
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Update on the Trump tariff basket, Intraday trend and 0DTEs, Risk premia portfolios for stagflation times
Global Quantitative Strategy	Robert Smith, PhD	BuzzMiner for Global News	Major Themes for Markets (Tariff Edition)
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Systematic Highlights	The Woods Are Lovely, Dark and Deep
Global Quantitative & Derivatives Strategy	Emma Wu	Global Equity Derivatives Strategy	Trading US Macro Events with Short-Dated Options
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Profiting from Tariffs, 0DTE vol skew and Evolving the timing model
Global FX Strategy	Antonin T Delair	Trump Social Media	The FX Impact
Global Quant Strategy	Ayub Hanif, PhD	Machine Learning Quant Equity Risk Factors	Can dynamic Machine Learning models be used to redefine traditional Equity Risk Factors?
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	How to get optimal equity factor exposure, Systematic short Rates vol: Swaptions or Treasury options?
Global FX Strategy	Antonin T Delair	FX Macro Quant	Changing dynamics between FX & equities
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Systematic Highlights	Market Regime Classification for QIS indices
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	QIS implementations of cross-asset views, short vol strategies & the January effect
Global Quantitative Strategy	Robert Smith, PhD	Global style seasonality	Cheap and Cheerful Stocks, Sectors and Countries for the New Year

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

List of 2024 reports with focus on systematic investing

Team	Lead AC	Title	Subtitle
Global Quantitative Strategy	Robert Smith, PhD	Big Data & AI Strategies	News themes from BuzzMiner (December 2024 update)
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Portfolio optimization with options	A method to weight historical scenarios
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Implications of the 2025 outlooks for QIS investors
Emerging Markets Strategy	Tales Padilha	EM Quant Series: Fair Value in EM Sovereign Credit	Introducing our new EM Sovereign Credit Fair Value
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	US elections effect on ARP performance and Puts vs defensive trend-following
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	0DTE options: The Day-of-the-week effect	Connecting vol premium with flows
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	0DTE options: The Day-of-the-week effect	Connecting vol premium with flows
Global FX Strategy	Antonin T Delair	FX Macro Quant	Pause macro drivers during elections
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	The right strikes for your SPX hedge, ARP when Fed is easing and FX Carry via options
Global Quantitative Strategy	Robert Smith, PhD	Investable AI	Earnings Call Analysis Using SmartBuzz – Solving Rational Inattention
Global Quantitative Strategy	Robert Smith, PhD	Investable AI	Using GenAI for Financial Analysis
Global Quantitative Strategy	Robert Smith, PhD	Investable AI	Global News impact on Equities (BuzzMiner, September 2024)
Global Quantitative & Derivatives Strategy	Arindam Sandilya	FX Volatility	Not for the faint-hearted
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Strategy	'Leaders and Followers' for a broader cross asset universe
Global Quantitative & Derivatives Strategy	Davide Silvestrini	Global Equity Derivatives	Systematic Dispersion Update and Trade Ideas
Global Quantitative & Derivatives Strategy	Juan Duran-Vara	FX Carry via Options	Deploy premium and sleep well at night
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Resilient option carry through dynamic strikes selection	An update
Global Cross Asset Strategy	Olivier Daviaud, PhD, CFA	Cross Asset Strategy	Where are we now? Market Regime Classification update
Emerging Markets Research	Saad Siddiqui	EM Credit Risk-Adjusted Carry	CRAC-ing Convergence: CRAC indices now in DataQuery and Bloomberg

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	TF in x-asset factors, ETF-flow sector rotation, 0DTEs seasonality and latest model views
Global FX Strategy	Antonin T Delair	FX Derivatives Research Note	Predicting and Trading the Volatility Surface
Global Quantitative Strategy	Robert Smith, PhD	Investable AI	Enhancing VIX Forecast Accuracy through Advanced Modeling Techniques and Feature Engineering with ChronoML
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Same-day options on SPX	0DTEs from a systematic trading perspective
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Timing equity factors, SPX Put ratio tail hedges, Pure cross-asset momentum and latest model views
Global FX Strategy	Antonin T Delair	FX Macro Quant	The clock is ticking for G10 carry
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Option carry, Pure single stocks statistical momentum, Commodity time spreads skewness, same day options on SPY
Global Quantitative & Derivatives Strategy	Thomas Salopek	Cross Asset Strategy	Market Regime Classification
Global FX Strategy	Lorenzo Ravagli, PhD	FX skew risk premium	Term structure effects and timing of the trades
Global FX Strategy	Antonin T Delair	The FX VRP conundrum	Long term vs short term Volatility Risk Premium
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Deviation from fundamental fair value revisited, Single stocks TF in factors, 0DTE options and economic releases and latest model views
Emerging Markets Research	Saad Siddiqui	EM Quant Series: Credit risk-adjusted FX carry	Adjusting for sovereign risk significantly boosts EM FX carry returns
Global Quantitative & Derivatives Strategy	Juan Duran-Vara	FX Derivatives	FX Carry via Options looks attractive
Global Quantitative & Derivatives Strategy	Emma Wu	Big Data and AI Strategies	Expanding Coverage of High-Frequency CFTC Futures Predictions
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Rates time spread skew, Trend-following in single stocks, Calendar effects short-dated options, Delta-hedged option risk profile and latest model views
Global Quantitative & Derivatives Strategy	Davide Silvestrini	European Equity Derivatives Outlook	Deep Dive on Index Call Strategies - Optimizing Naked and Delta-Hedged Systematic Short Calls and Tactical Hedging of CTA Reversals
Global FX Strategy	Antonin T Delair	FX Macro Quant Model Summary	If US growth stays, carry (and USD) can still deliver
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Beyond the vol premium, Optimal skewness strategy, Timing long-only low vol equity and latest model views
Emerging Markets Strategy	Tales Padilha	EM Quant Series	The Trend Is Your Friend: Assessing momentum-based strategies in EM Fixed Income
Global Quantitative Strategy	Robert Smith, PhD	Big Data & AI Strategies	News themes from BuzzMiner (March 2024 update)
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Big Data and AI Strategies	Deep Learning Portfolio Construction

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Juan Duran-Vara	EM FX Vol Carry Portfolios	Evaluating EM FX Vol Carry portfolios; a great addition to a cross-asset risk-premia portfolio
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Performance review, rates fair value and momentum, EUR swaptions risk premium, residual skewness as signal and latest model views
Emerging Markets Strategy	Tales Padilha	EM Quant Series	Monitoring Fair Value in Local Bond Yields: Introducing our new fair value monitoring tool for 10y, 5y and GBI-EM yields
Global Quantitative Strategy	Robert Smith, PhD	Big Data & AI Strategies	VXY Mariner Edition – February 2024
Global Quantitative Strategy	Robert Smith, PhD	News Influence on Equity Markets	Theme Betas of Global News Trends (BuzzMiner)
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	MOC Imbalance as a Signal	An extension to NDX, RTY and BTIC
Global Quant Strategy	Ayub Hanif, PhD	The Evolution of Quant Factor Investing	Using Machine Learning models for feature engineering and systematic stock selection
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Catching the (third) moment	Harvesting the risk premia in left skewness
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Trade at Settlement Futures Markets	Signaling value in Commodities Futures
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Investable AI Highlights	Update - January 2024
Global Quantitative & Derivatives Strategy	Emma Wu	Investable AI for Volatility Trading	Deep Learning Model for Cross Asset Volatility Strategies
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	2023 in the rear view mirror and latest model views
Global Quantitative & Derivatives Strategy	Emma Wu	Cross Asset Volatility	Machine Learning Trading Model Performance Update 4Q23

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

List of 2023 reports with focus on systematic investing

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Emma Wu	0DTE options	A postmortem on the recent intraday sell off
Global Asset Strategy	Thomas Salopek	Cross Asset Strategy Quarterly	3Y to 5Y CMAs and Portfolio Construction: Themes for the Medium-Term
Global FX Strategy	Antonin T Delair	FX Macro Quant	Finally some G10 value
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	JPM Research 2024 outlooks and XRP strategies, zooming in on put ratios and latest model views
Global FX Strategy	Lorenzo Ravagli, PhD	FX Derivatives	May I suggest a vol spread or two?
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset CTA Activity Update	Positioning Headwinds Replace Tailwinds
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	0DTE options	How to distinguish HFT from retail orders
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data & AI Strategies	History of AI Slide Deck
Global FX Strategy	Meera Chandan	FX Macro Quant 2024 Outlook	The big trades are in G10 valuation gaps
Global Quantitative & Derivatives Strategy	Libin Cheng	Cross Asset Volatility	Relative value opportunities for macro hedging
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	Macro Risk News Report (Mariner) November 2023
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	VXY Mariner Edition, November 2023
Global Quantitative & Derivatives Strategy	Arindam Sandilya	FX Volatility	Not going anywhere fast
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Opportunities in synthetic defensive baskets , revisiting the quest for pure equity factor exposure, the choice of a delta-hedging scheme and latest model views
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset CTA Activity Update	Estimated $92B of Buying in US Treasury Futures over Next Month
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data & AI Strategies	SmartBuzz Updates v3.1
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset CTA Activity Update	Positioning estimates from high frequency data
Global Asset Strategy	Thomas Salopek	Cross Asset Strategy	Spotlight on Central Banks
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	BuzzMiner update October 2023
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	J.P.Morgan digest on risk premia strategies	Summary of Q3'23 research reports on systematic investing
Global FX Strategy	Antonin T Delair	FX Macro Quant	Rates based strategies weaken in favor of other fundamentals
Global FX Strategy	Ladislav Jankovic	FX Derivatives Research Note	Mahalanobis distance measure proves forward looking for FX vol

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data & AI Strategies	Time Series Analysis in the Multi-verse with JPM's ChronoML
Global FX Strategy	Antonin T Delair	FX Macro Quant Model Summary	Stuck between yields and valuations
Global Quantitative & Derivatives Strategy	Thomas Salopek	Cross Asset Strategy	Risk on/Risk off signals with the Boruta + PCA + GLM pipeline
Global Quantitative & Derivatives Strategy	Emma Wu	Intraday Price Patterns from Option Gamma Imbalances	Timing of delta hedge; Signaling value from Gamma profile
Global FX Strategy	Antonin T Delair	FX Macro Quant Model Summary	Strong rates RV forces pushing USD despite benign growth environment
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	How close to realized vol should implied vol trade?	
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data & AI Strategies	Generative AI for Investments Deck
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Research	VXY Mariner Edition
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data & AI Strategies	Generative AI for Investments Deck
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Options P&L Attribution	Theory and applications
Global FX Strategy	Ladislav Jankovic	FXO event risk preview	FXO markets price BoJ as the key event of the week. Is the rest underpriced?
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Revisiting retail order flow as a signal for US equities	New algorithm and intraday dynamics
Global Quantitative & Derivatives Strategy	Thomas Salopek	Cross Asset Strategy	Discrete risk-based rules for effectively hedging portfolios
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Big Data and AI Strategies	Return Predictability by AI Algorithms
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	SmartBuzz – Keeping an Eye on AI
Global FX Strategy	Antonin T Delair	FX Macro Quant Model Summary	Carry losses on the shorts but remain in positive territory over the rolling month
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	When underhedging FX risk is optimal	Using the FX/Asset correlation as an input
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	Macro Risk News Report (Mariner) update – July 2023
Global Quantitative & Derivatives Strategy	Libin Cheng	Global Equity Derivatives	Systematic Dispersion Update
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Payer swaptions as duration proxy, the signal gist of overnight trading and latest model views
Global Quantitative & Derivatives Strategy	Erik Rubingh	Quantitative Perspectives on Cross-Asset Risk Premia	Quantifying the risk profile of an option butterfly, Timing low vol equity and latest model views
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	The Signal Gist of Overnight Trading	Profiting from the Information Content of Overnight and Daytime Equity Index Returns
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data & AI Strategies	Using Generative AI for investing, including effective ChatGPT prompts
Global FX Strategy	Meera Chandan	FX Macro Quant Mid-year Outlook	Still some juice left in carry before the end
Global FX Strategy	Arindam Sandilya	FX Derivatives	Too low for late-cycle conditions
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Risk Premia Highlights	Sensitivities to equity and bonds tails, asymmetric trend-following ARP hedge and the best option strike/maturity combo

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	BuzzMiner now identifies the Most Mentioned Entities for each theme
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	Macro Risk News Report (Mariner) update - Jan 2023
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Drivers of Market-on-Close(MOC) Imbalance as a Signal	An Analysis of Liquidity Conditions and Crowdedness
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Locking in rates vol carry, overnight signal strategy vs. intraday momentum and latest model views
Global Quantitative and Derivatives Strategy	Olivier Daviaud, PhD, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Timing systematic long rates vol, fundamental rates value and latest model views
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Fundamental Rates Value Strategies	Harnessing macro-economic data to trade yield deviations from fair values
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Investable AI for Volatility Trading	Introducing the New ETF RV Volatility Model
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	Macro Risk News Report - VXY Mariner Edition
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Performance review, factor trend-following applications: pure equity and cross-asset factors and latest model views
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data and AI Strategies	Summary of J.P. Morgan research and industry developments in 2H22
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	0DTE options	Retail participation and Volmageddon risk quantified
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data and AI Strategies	Reducing Source Bias, Niche Neutralization, and Dynamic Subtopic Modeling
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data and AI Strategies	NLP for Investing
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	0DTE options	A breakdown of common structures
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Swaptions Carry Calculation, The Value in Equity Value and latest model views
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data and AI Strategies	Reviewing Company Results Daily with SmartBuzz for Sentiment and Themes
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia for fixed income enhancement

List of 2022 reports with focus on systematic investing

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	BuzzMiner Enhancements	Adjusting for Niche Themes and Source Bias
Global Quantitative & Derivatives Strategy	Thomas Salopek	Cross Asset Strategy	Capital Market Assumptions: Robust modelling of 3Y to 5Y expected returns – Bayesian vs Traditional
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset Volatility	Machine Learning Based Trade Recommendations
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, 2023 Outlook and risk premia, vol strategies, carry strategies and basis-momentum
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	High Frequency Predictions of CFTC Futures Positioning Data Follow-up
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	How close to realised should implied vol trade	Revisiting P&L attribution for SPX options
Global Quantitative & Derivatives Strategy	Marko Kolanovic, PhD	2023 Equity Derivatives Outlook	Volatility Forecasts and Trade Ideas
Global Quantitative & Derivatives Strategy	Libin Cheng	Big Data and AI Strategies	High Frequency Predictions of CFTC Futures Positioning Data Follow-up
Fixed Income Strategy	Srini Ramaswamy	Interest Rate Derivatives 2023 Outlook	From tiger's roar to rabbit's foot
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Risk Premia Highlights	Systematic rates volatility in 2022 and rates carry prospects
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, Systematic Rates Vol in 2022, prospects for Rates Carry and latest model views
Global Quantitative & Derivatives Strategy	Lorenzo Ravalgi, PhD	FX Derivatives	Vols are stunned, not dead
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	JPM FX - Derivatives Chartpack Notes	JPY and GBP probability distributions - sell 2yr 25delta USD/JPY risk-reversal
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Risk Premia Highlights	Rates Value strategies and introducing the thematic publications on applied portfolio construction
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, profiting from the reversal in term spread and latest model views
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review and exploring value strategies in rates
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, update on Variation in Hedging Pressure and Crypto Trend-Following
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Performance review, trend-following revival, analysis of vol selling and latest model views
Global FX Strategy	Antonin T Delair	FX Macro Quant 2H Outlook	Carry is back, but doesn't protect against a recession
Global Markets Strategy	Pedro Martins Junior	Key Trades and Risks	Three FAQs and Our Views
Global Quantitative & Derivatives Strategy	Emma Wu	Big Data and AI Strategies	Machine Learning gamma trading models for EUR and JPY x-vol
Global Markets Strategy	Pedro Martins Junior	EM Lighthouse: Face the EM Style Factors	EM Equity Strategy Factor Investing
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Analyzing YTD performance of delta-hedged puts and incorporating macro-economic sensitivities in the pure equity factors
Global FX Strategy	Meera Chandan	A faster TEAM*	Using market-based signals in a multi-factor approach to FX
Global FX Strategy	Juan Duran-Vara	JPM FX - Derivatives Chartpack Notes	CZK this out! Fading the elevated skews in CEE | Fade vol convexity via Vol/Var swap RVs on AUD/USD
Global Quantitative & Derivatives Strategy	Thomas Salopek	Cross Asset Strategy	What if the mean reverts?

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Robert Smith	Big Data and AI Strategies	Machine tags for thematic investing (SmartTag)
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, impact of rates and inflation on equity factors and latest model views
Credit Derivative and Index Reserarch	Saul Doctor	A Switch in Time Saves Nine	Credit sector rotation strategies
Global Quantitative & Derivatives Strategy	Lorenzo Ravalgi, PhD	JPM FX - Derivatives Chartpack Notes	USD/JPY Var vs. Vol swap opportunity | RV sentiment model buying vol on CNH
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	MOC Imbalance as a Signal	Performance Update
Global Quantitative & Derivatives Strategy	Emma Wu	Big Data and AI Strategies	Robust Out-of-sample Performance Makes a Strong Case for Machine Learning FX Trading Models
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Update on pure defensive equity factors and defensive risk premia portfolios and profiting from the positioning dynamics
Global FX Strategy	Ladislav Jankovic	FX Derivatives Research Note	TARN/Target redemption forwards as a yield enhancing product - a primer
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Profiting from Positioning Dynamics in Commodity Futures	Hedgers vs. Speculators
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	What Drives Alpha in Intraday Momentum?	A Case Study in HYG
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, trend-following strategies for inflation protection and latest model views
Global Quantitative and Derivatives Strategy	Haoshun Liu	CSI 500 Futures Implied Funding	Supply and demand dynamics review, Interpretable forecasting models using XGBoost and SHAP
Global Quantitative and Derivatives Strategy	Peng Cheng, CFA	Cross Asset Volatility	Carr-Wu Credit Equity Relative Value Opportunities
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Performance review and tailored systematic strategies for rates-up protection
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Impact of Crowding on Intraday Momentum	A Market Structure Analysis
Global FX Strategy	Juan Duran-Vara	Macro Factors and FX Vols	
Global Quantitative & Derivatives Strategy	Thomas Salopek	Cross Asset Strategy	Rising rates and the 60/40

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

List of 2021 reports with focus on systematic investing

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, oil replication systematic strategies and latest model views
European Credit Strategy and Derivative Research	Shivam Ghosh	Factor Tracker	Credit Factor funds, J.P.Morgan's credit smart beta ESG index and 3Q performance update
Global Quantitative & Derivatives Strategy	Tony SK Lee	Asia Pacific Equity Derivatives Highlights	Japan gamma positioning review, NKY FVA, Adaptive Regime Compass Oct-21 update, Volatility positioning and dynamics
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Big Data and AI Strategies	Do Retail Investors Harvest Tax Losses?
Global Quantitative & Derivatives Strategy	Haoshun Liu	AI and Big Data Approach to Thematic Investing	Measuring the topicality of LDP election candidates' policy proposal; Trade ideas for the Japan elections
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset Volatility	Relative Value Volatility Model Performance Update
Global Quantitative & Derivatives Strategy	Tony SK Lee	Asia Pacific Equity Derivatives Highlights	Introducing ARC model for Australia; AS51 hedging strategies
Global Quantitative & Derivatives Strategy	Robert Smith	Thematic Equity Factors From the News Using SmartBuzz 2.0	News-Based 'Sustainable Dividends' Equity Factor Using Only NLP
Global Quantitative & Derivatives Strategy	Haoshun Liu	AI and Big Data Approach to Thematic Investing	Assessing Japan digital transformation beneficiaries following the establishment of the Digital Agency
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	J.P.Morgan digest on risk premia strategies	Summary of Q2'21 research reports on systematic investing
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli	JPM FX - Derivatives Chartpack Notes	Systematic FX vol models - a mid-summer overview of the latest signals
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, update on defensive risk premia performance and latest model views
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	To YOLO or not to YOLO	A Comparative Analysis of Retail Sentiment and Order Flow
Global Quantitative & Derivatives Strategy	Tony SK Lee	Introducing the Adaptive Regime Compass	Measuring Equity Market Similarities with ML Algorithms
Global FX Strategy	Ladislav Jankovic	FX Derivatives Research Note	Double no-touches go quantum, the first look
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	JPM FX - Derivatives Chartpack Notes	Long Vega and Theta-collecting constructs best supported by vol surfaces setup
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Risk premia performance review, trend-following in factors and latest model views
Global FX Strategy	Ladislav Jankovic	FX Derivatives Research Note	Distance measure for quantifying market regimes for FX vol trading
European Credit Strategy and Derivative Research	Shivam Ghosh	Factor Tracker	2Q performance and H2 Outlook
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Global Equity Derivatives	Optimal Variance Replicating Portfolio
Global FX Strategy	Meera Chandan	FX Macro Quant	A neutralization across most signals
Global FX Strategy	Meera Chandan	T.E.A.M. 2.0*	A time-series multi-factor approach to FX
Global Quantitative & Derivatives Strategy	Thomas J Murphy	Do Retail Traders Move Commodity Markets?	Impacts of Retail Order Flow on Commodity Momentum Strategies

Team	Lead AC	Title	Subtitle
Global FX Strategy	Meera Chandan	FX Macro Quant Mid-year Outlook	Low yields and little value meet neutralizing growth momentum
Global FX Strategy	Arindam Sandilya	FX Derivatives - Midyear drama	FX Derivatives: Hawkish Fed supportive of higher FX vol
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Disentangling momentum and carry in commodities, structural breaks in risk premia and latest model views
Chief Global Markets Strategist	Marko Kolanovic, PhD	Thematic Investing Handbook	Global Outlook and Key Themes
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	JPM FX - Derivatives Chartpack Notes	RV of OTM strangles as sentiment indicators remain in risk-loving mode
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Retail Order Flow in US Equity Options	Case Studies and Signal Value
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Enhancing active ETF performance, cluster portfolio and tail risk parity updates, and latest model views
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset Volatility	Systematically monetize bond/equity correlation premium via vanilla options
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset Volatility	What can option implied skew tell us about bond equity correlation?
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Quarterly review, Credit value, Synthetic defensive long vol and latest model views
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data and AI Strategies	Assessing a range of NLP Transformer Models such as BERT, Electra, Funnel, GPT2, MPNet and variants
Global Quantitative & Derivatives Strategy	Khuram Chaudhry	ESG - Environmental, Social & Governance Investing	Introducing the 'Human Capital Factor'
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Big Data and AI Strategies	Tracking Retail Activity in the US Equity Options Market
Global Quantitative & Derivatives Strategy	Khuram Chaudhry	ESGQ	ESG Investing - short term vulnerabilities, long term opportunities
European Credit Strategy and Derivative Research	Shivam Ghosh	Virtue isn't the only reward	Active management under ESG constraints
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Pure ESG factors, new research on risk premia and latest model views
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Hedging rising inflation, trend-following system review, Bitcoin trend-following backtest and latest model views
European Credit Strategy and Derivative Research	Shivam Ghosh	Do Good; Get Paid	Estimating ESG Alpha in European Credit
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Risk Premia Investing in the Age of Machines	Investable Strategies with Machine Learning Asset Pricing Models for Global Equities
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Risk Premia Portfolio Construction via Uncorrelated Clusters	Introducing Autoencoders as an extension of Principal Components

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Team	Lead AC	Title	Subtitle
Chief Global Markets Strategist	Marko Kolanovic, PhD	Market and Volatility Commentary	Outlook remains positive - buy the dip; where are the bubbles and COVID-Green risk
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Reviewing 2020 performance, a note on January effect and latest model views
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	A (Sub)Penny Saved	Tracking Retail Trading Activity in US Equities and ETFs
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Inflation Expectation Tracking Equity Baskets	A Pure Equity Factor Approach

Dobromir Tzotchev, PhD AC
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Background report

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Yangyang Hou	Equity Volatility Strategy	What to Long, Short, and When (Part I)
Global FX Strategy	Antonin T Delair	Rates Macro Quant	Rates 'real' carry with a duration overlay
Global FX Strategy	Antonin T Delair	Rates Macro Quant	Swaps vs Bond Futures: the carry use-case
Global Quantitative Derivatives Strategy	Yangyang Hou	Equity Volatility Strategy	Thinking Outside the Box for Tail Trading Part-2
Global Quantitative Derivatives Strategy	Yangyang Hou	Equity Volatility Strategy	Thinking Outside the Box for Tail Trading
Global Quantitative & Derivatives Strategy	Emma Wu	Global Equity Derivatives Strategy	Trading US Macro Events with Short-Dated Options
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Portfolio optimization with options	A method to weight historical scenarios
Global Cross Asset Strategy	Thomas Salopek	Cross-Asset Systematic Strategy	CMAs, Portfolio Construction, Regimes, and the coming Policy impacts
Global Cross Asset Strategy	Thomas Salopek	Cross-Asset Systematic Strategy	Dynamically switching between Individual and Network Momentum
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Strategy	Combining Positioning Dynamics and Price Momentum in Cross Asset Futures
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Strategy	Quant approach to "Leaders and Followers"
Global Quantitative & Derivatives Strategy	Juan Duran-Vara	FX Carry via Options	Deploy premium and sleep well at night
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Strategy	Market Regime Classification
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	FX skew risk premium	Term structure effects and timing of the trades
Global Quantitative & Derivatives Strategy	Emma Wu	Big Data and AI Strategies	Expanding Coverage of High-Frequency CFTC Futures Predictions
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	Option risk profiles and dynamic strike selection	An update on defensive strategies
Global Quantitative & Derivatives Strategy	Erik Rubingh, CFA	Catching the (third) moment	Harvesting the risk premia in left skewness
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Strategy	Risk on/Risk off signals with the Boruta + PCA + GLM pipeline
Global Cross Asset Strategy	Thomas Salopek	Cross Asset Strategy	Risk premia from pairs to portfolios
Credit Derivative and Index Research	Lovjit S Thukral, PhD	Systematic Strategies in Corporate Credit	Volume 1: Relative Value in USD IG: From theory to practice
Global Cross Asset Strategy	Federico Manicardi	Cross Asset Strategy	With a mixed macro environment and range-bound markets, we upgraded our PCA tool to screen dislocations
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	FX Derivatives	Between a rock and a hard place
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset Volatility	Updates on Macro Credit Equity Relative Value
Global Quantitative & Derivatives Strategy	Emma Wu	Trade at Settlement Futures Markets	Market Liquidity and Information Contents
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	Reviewing Company Results Daily with SmartBuzz for Sentiment and Themes
Global Quantitative Strategy	Khuram Chaudhary	European Factor Reference Book	What does and doesn't work for European quants
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data & AI Strategies	Macro Risk News Report (Mariner) update - Jan 2023
Global Quantitative & Derivatives Strategy	Erik Rubingh	Basis Momentum	Another profitable variant of the momentum phenomenon
Global Quantitative & Derivatives Strategy	Olivier Daviaud, PhD, CFA	How close to realised should implied vol trade?	Revisiting P&L attribution for vanilla SPX options
Global Quantitative & Derivatives Strategy	Emma Wu	Systematic late-cycle hedging with FX Options	A Machine Learning based strategy

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	Options Skew Risk Premium	How to measure and harvest it
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Retail order flow in US equity options	Signal value for stock performance
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	VIX Technical Update	Information Content in Trade at Settlement (TAS) Futures
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Big Data and AI Strategies	Machine tags for thematic investing (SmartTag)
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data & AI Strategies	State of the Art NLP Tools with Practical Financial Applications such as Thematics, Summarisation, Q&A
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	Building FX Vol portfolios	Practical takeaways for hedging and trading purposes
Global Credit Research	Shivam Ghosh	ESG Investing	
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data and AI Strategies	SmartBuzz for Trading Thematics in Earnings Call Transcripts and the News
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Quantitative Perspectives on Cross-Asset Risk Premia	Oil replication systematic strategies and latest model views
Global Quantitative & Derivatives Strategy	Robert Smith, PhD	Thematic Equity Factors From the News Using SmartBuzz 2.0	News-Based 'Sustainable Dividends' Equity Factor Using Only NLP
Global Quantitative & Derivatives Strategy	Berowne Hlavaty	Big Data & AI Strategies	Assessing a range of NLP Transformer Models such as BERT, Electra, Funnel, GPT2, MPNet and variants
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli	Trading vol-of-vol risk premium	Harvesting smile convexity across asset classes
Global Quantitative & Derivatives Strategy	Haoshun Liu	AI and Big Data Approach to Thematic Investing	Structural digitalization trends in cloud computing, telehealth, video gaming and cybersecurity
Global FX Strategy	Meera Chandan	T.E.A.M.*	Introducing a multi-factor approach to FX
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Follow the Robinhood Money	Buying Behavior and Market Impacts of Individual Traders
Global Quantitative & Derivatives Strategy	Khuram Chaudhry	ESGQ	Is ESG performance simply a measure of long Technology & short Energy?
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Defensive Risk Premia	Systematic Strategies for the Risk-Off Times
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset Volatility	From Relative Value Signals to Optimal Portfolio Weights
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Custom Performance Attribution based on Portfolio Holdings	Decomposing Risk and Return Drivers via Factor-Mimicking Portfolios
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	The quest for pure equity factor exposure	How to eliminate the unwanted biases in equity factors?
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Cross Asset Volatility	Optimal Portfolio Construction - Beyond Risk Parity
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli	JPM FX - Derivatives Chartpack Notes	Assessing the impact of rates correlations on FX vols
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	Timing FX short-vol strategies	A systematic approach
European Credit Strategy and Derivative Research	Shivam Ghosh	Fact or Fiction	Investigating Factors in Corporate Credit
Global Quantitative & Derivatives Strategy	Robert Smith	Tracking Thematics in Equities	NLP with Phrase Embedding and Clustering for Robust Theme Tracking ('Smart Buzz')
Global FX Strategy	Meera Chandan	J.P. Morgan FX single-factor style frameworks	
Global Quantitative & Derivatives Strategy	Lorenzo Ravagli, PhD	Optimal option delta-hedging	Uncovering the link between mean-reversion and options strategies across markets
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Market-neutral carry strategies	Harvesting carry without market risk

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Team	Lead AC	Title	Subtitle
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	A Quantitative Framework for Cross Asset Style Timing	Machine Learning, Macro and Time-Series models providing views for Portfolio Tilting
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Global Equity Derivatives	A simple framework for credit-equity relative value trading
Global Quantitative and Derivatives Strategy	Peng Cheng, CFA	Global Equity Derivatives	A simple framework for credit-equity relative value trading
Global Quantitative & Derivatives Strategy	Peng Cheng, CFA	Global Equity Derivatives	US ETF Relative Value Volatility Model
Global Quantitative & Derivatives Strategy	Khuram Chaudhry	ESG - Environmental, Social & Governance Investing	Launching JPM's ESGQ - A Quantitative ESG Metric for stock selection models.
Global Quantitative & Derivatives Strategy	Dobromir Tzotchev, PhD	Designing robust trend-following system	Behind the scenes of trend-following
Credit Derivative and Index Research	Saul Doctor	CD Player	Market Themes and Relative Value Trade Ideas in the European Credit Derivatives Market
Quantitative and Derivatives Strategy	Marko Kolanovic, PhD	Equity Risk Premia Strategies	Risk Factor Approach to Portfolio Management
Global Equity Derivatives & Delta One Strategy	Marko Kolanovic, PhD	Market Impact of Derivatives Hedging	How Index Options and Variance Swap Hedging Impact the Market Near the End of a Trading Day

Disclosures

Analyst Certification: The Research Analyst(s) denoted by an "AC" on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an "AC" on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst's personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst's analysis was made in good faith and that the views reflect the Research Analyst's own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting https://www.jpmm.com/research/disclosures, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting https://www.jpmm.com/research/disclosures, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:
J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, https://www.jpmorganmarkets.com.

J.P. Morgan Equity Research Ratings Distribution, as of April 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	51%	37%	12%
IB clients**	83%	79%	74%
JPMS Equity Research Coverage*	49%	39%	13%
IB clients**	94%	93%	85%

***Please note that the percentages may not add to 100% because of rounding.**
****Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.**
For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at http://www.jpmorganmarkets.com, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, http://www.jpmorganmarkets.com. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:
A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of http://www.jpmorganmarkets.com where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to UK MIFID Research Unbundling exemption for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see https://www.jpmorgan.com/disclosures/cryptoasset-disclosure.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit https://www.theocc.com/components/docs/riskstoc.pdf for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:
Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting https://www.jpmm.com/research/disclosures. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: J.P. Morgan Australia - Research Independence Policy.

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC).

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: http://www.jpmipl.com. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Prasanna Bandal; prasanna.bandal@jpmchase.com; +912261575159. Grievance Officer: Ramprasadh K, jpmipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit Terms and Conditions and Most Important Terms and Conditions (MITC). The annual Compliance audit report is available at http://www.jpmipl.com/#research.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission.

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of “accredited investors,” “expert investors” or “institutional investors,” as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange (“Non-Taiwan Listed Securities”), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the “Important Disclosures” in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission, and its registered address is 3rd Floor, 20 North Sathorn Road, Silom, Bangrak, Bangkok 10500.

UK: Research is produced in the UK by J.P. Morgan Securities plc (“JPMS plc”) which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited (“JPMML Ltd”) which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 (“the FPO”); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA’s policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: J.P. Morgan EMEA - Research Independence Policy.

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: https://www.jpmorgan.com/disclosures/email

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to msci.com/disclaimer

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised April 04, 2026.

Dobromir Tzotchev, PhD [AC]
(44-20) 7134-5331
dobromir.tzotchev@jpmorgan.com

Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 15 May 2026 02:19 PM BST Disseminated 15 May 2026 03:09 PM BST